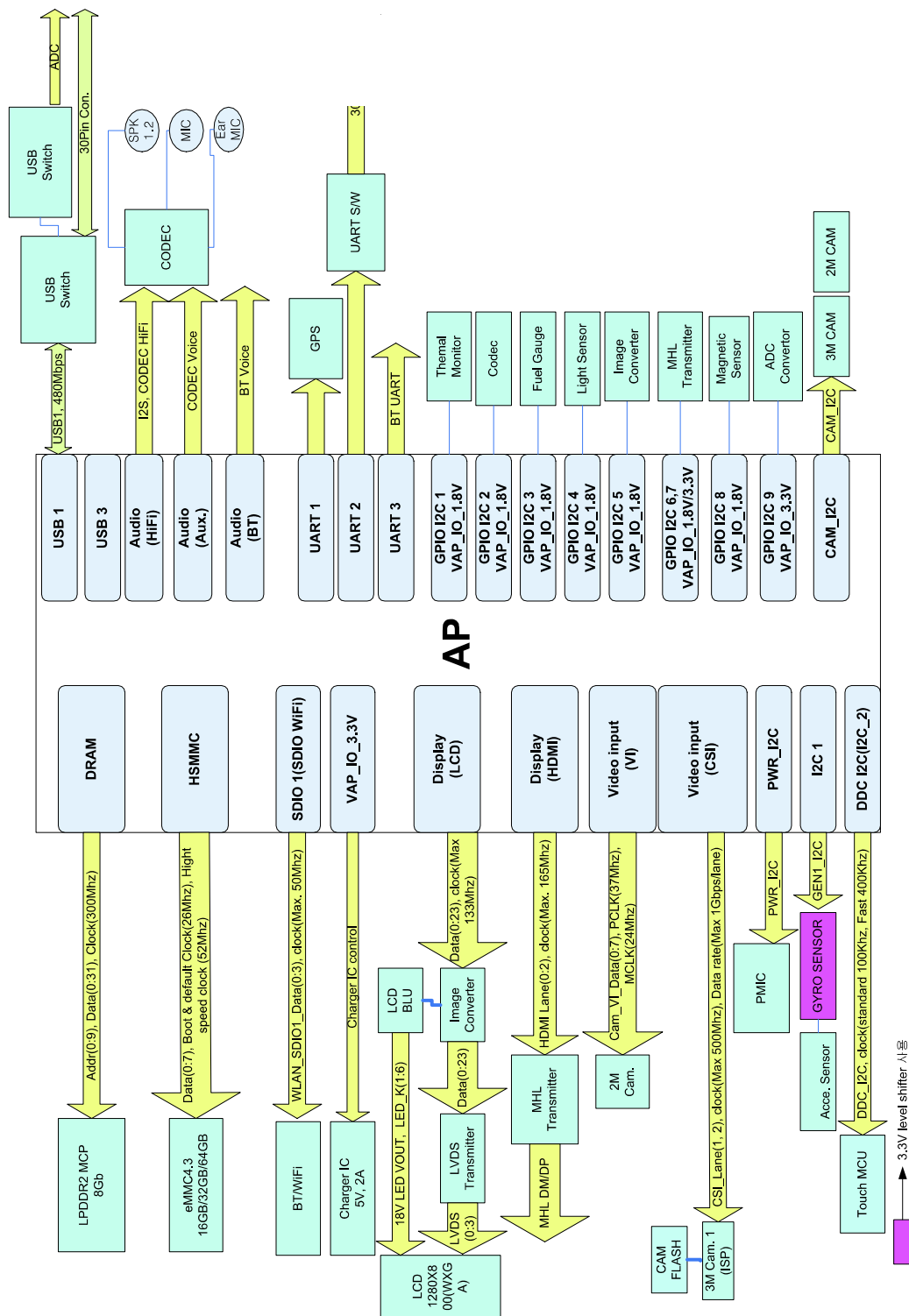


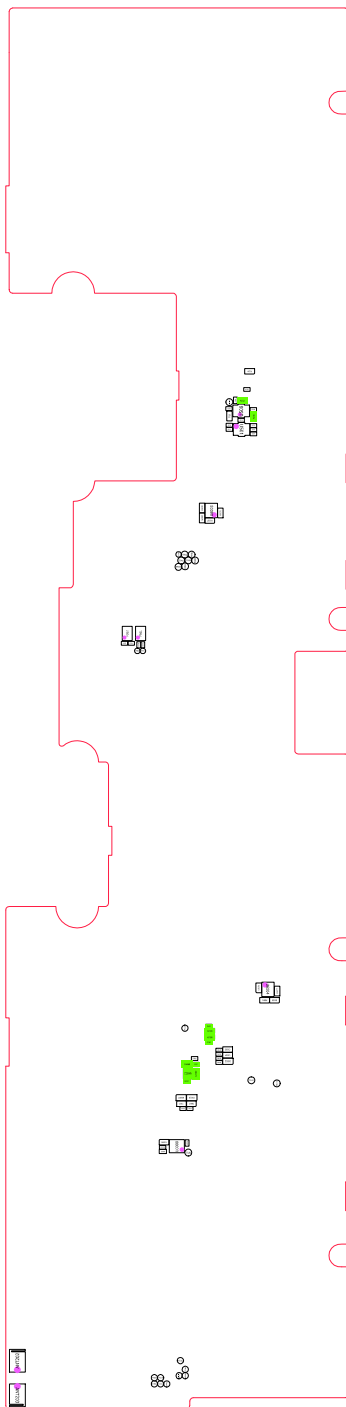
8.3 级维修

8-1.框图

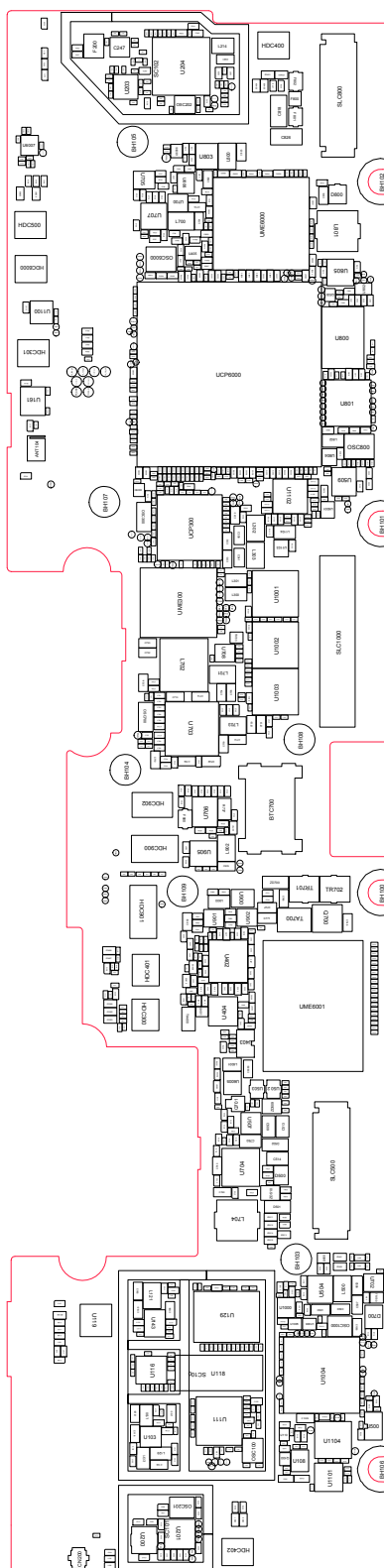


8-2.主板图

8-2-1.顶部

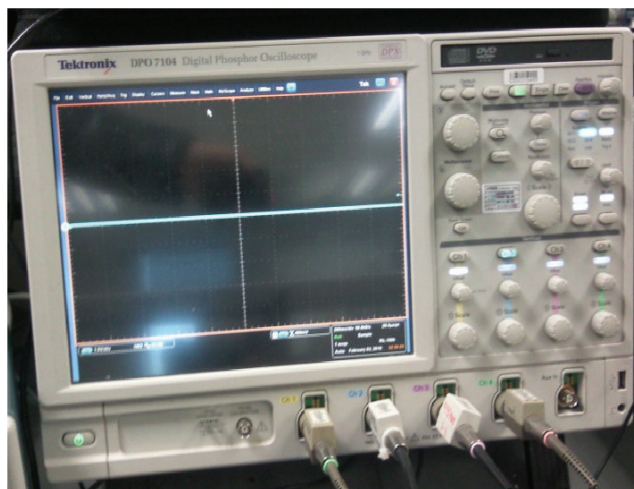


8-2-2.底部



8-3.故障排除流程图

设备



示波器



数字万用表

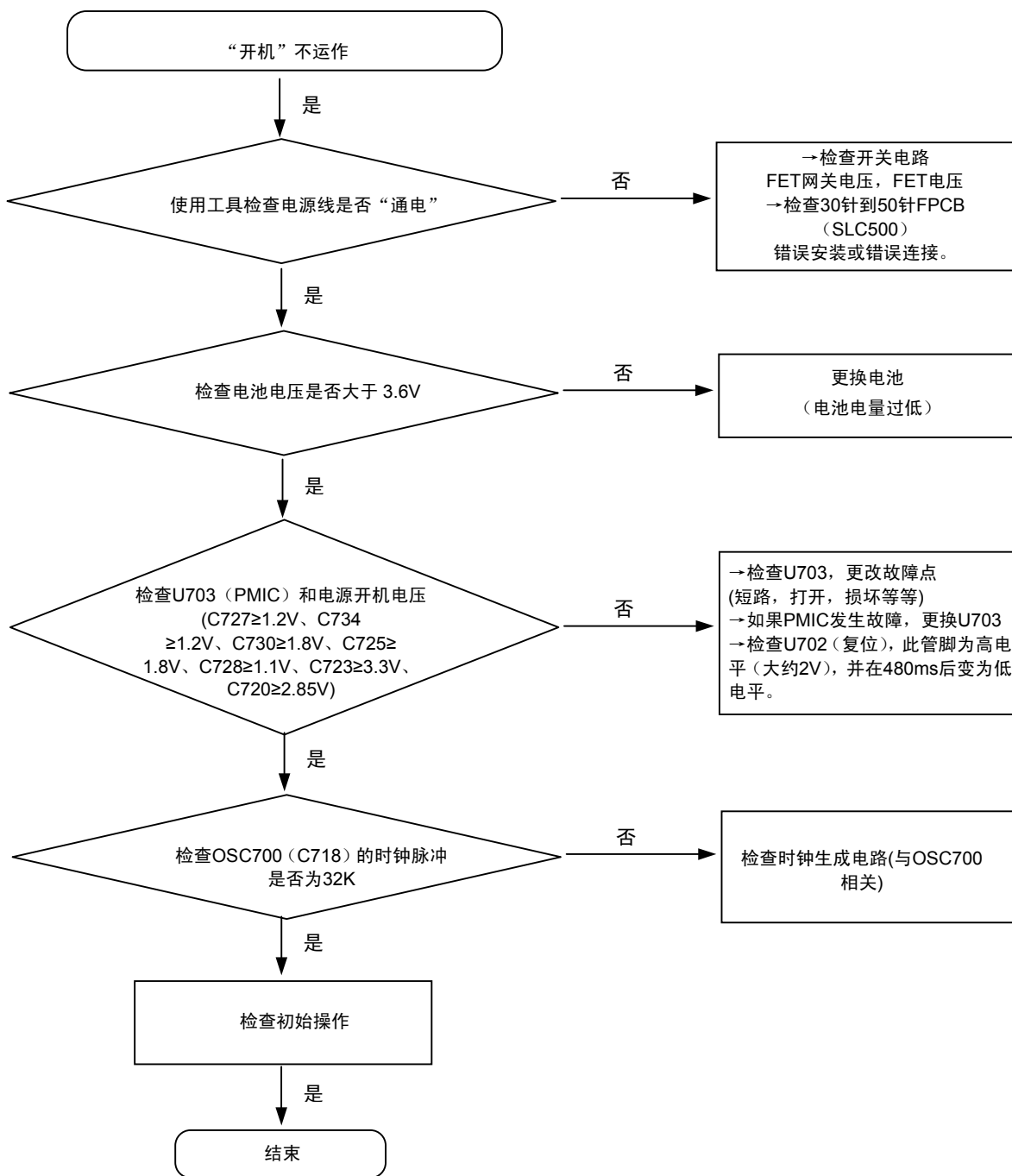


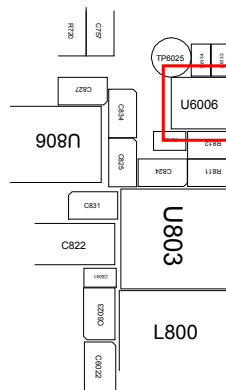
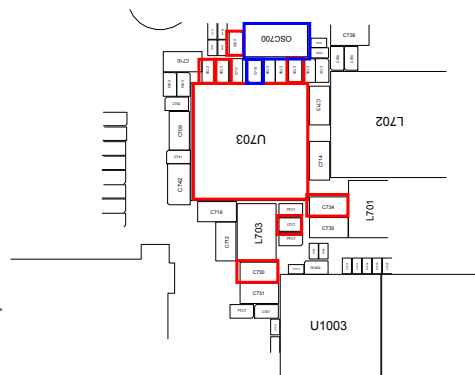
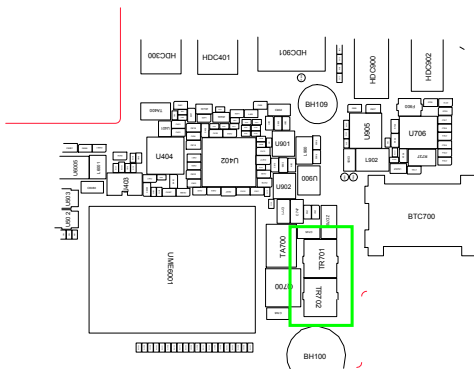
电源供电器



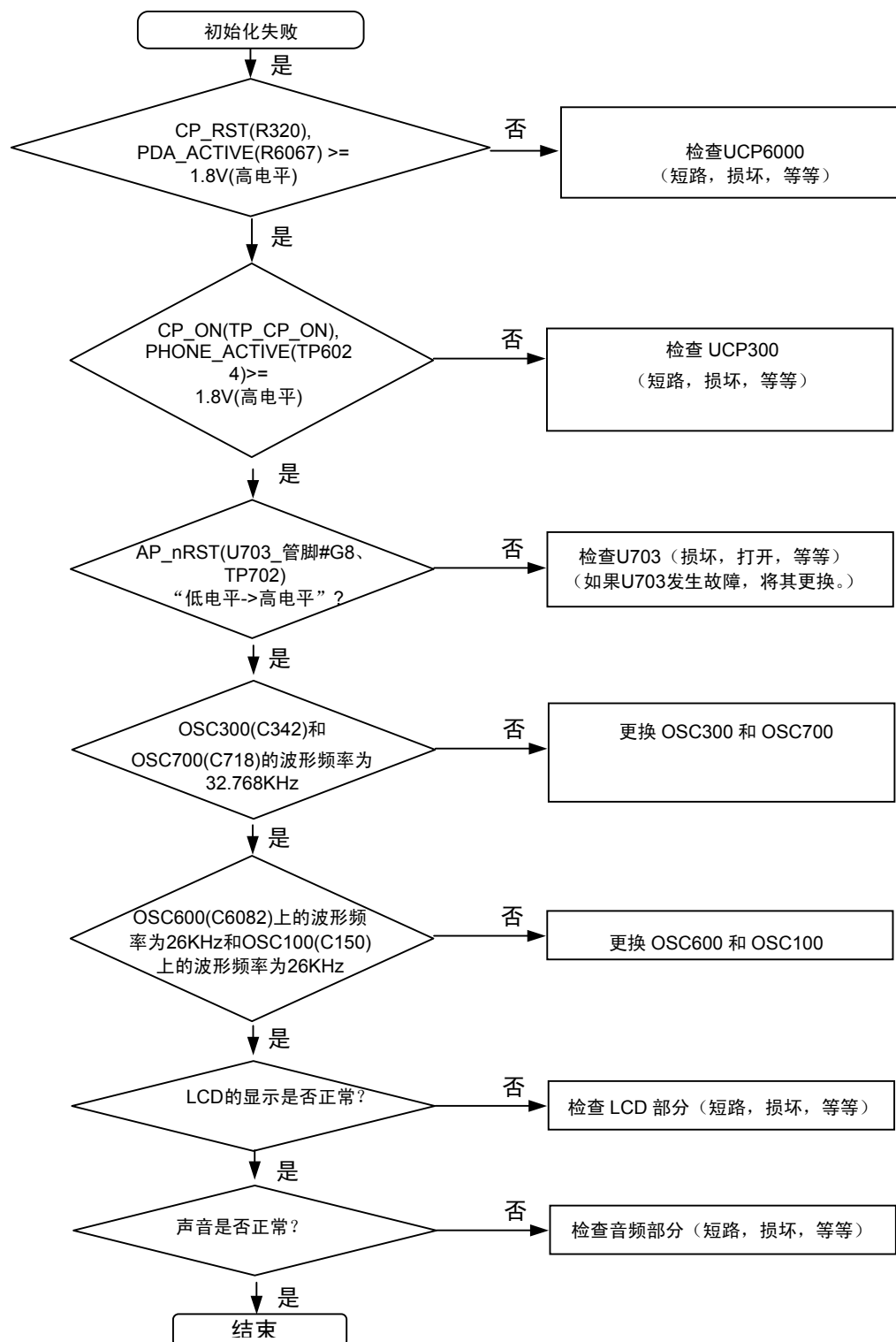
+字螺丝刀，小镊子

8-3-1 开机

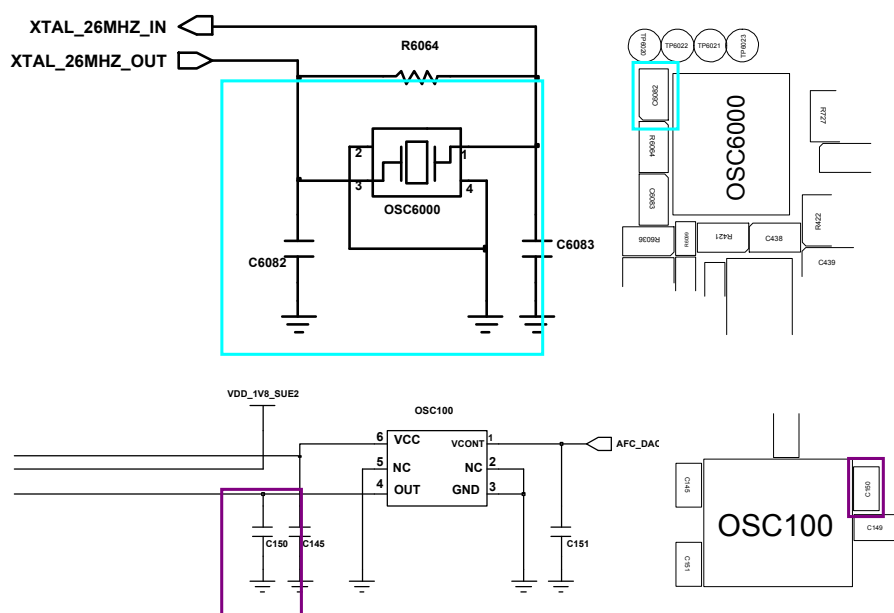




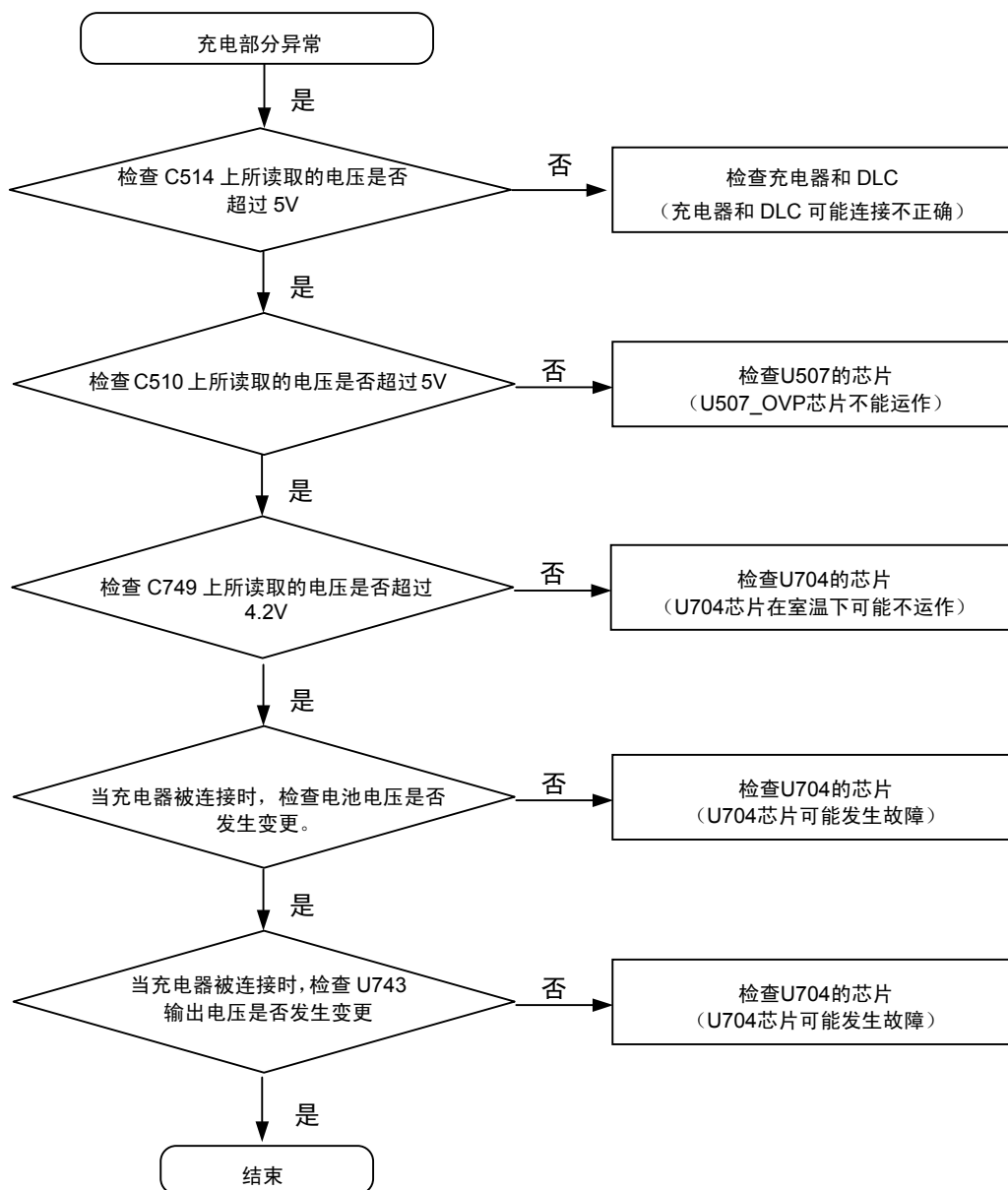
8-3-2.初始化

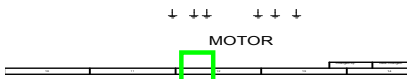
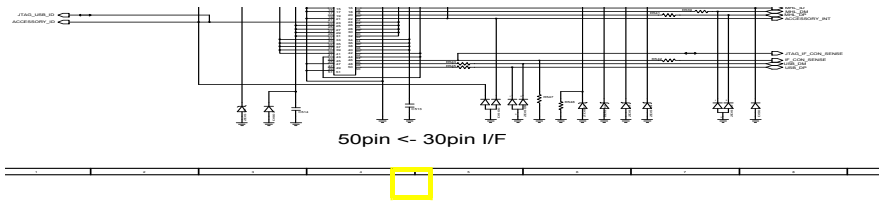




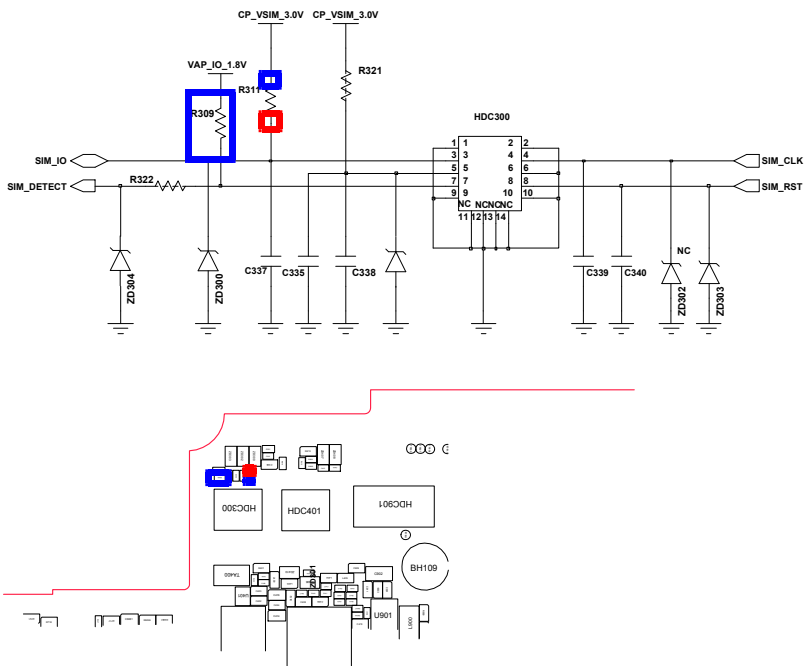
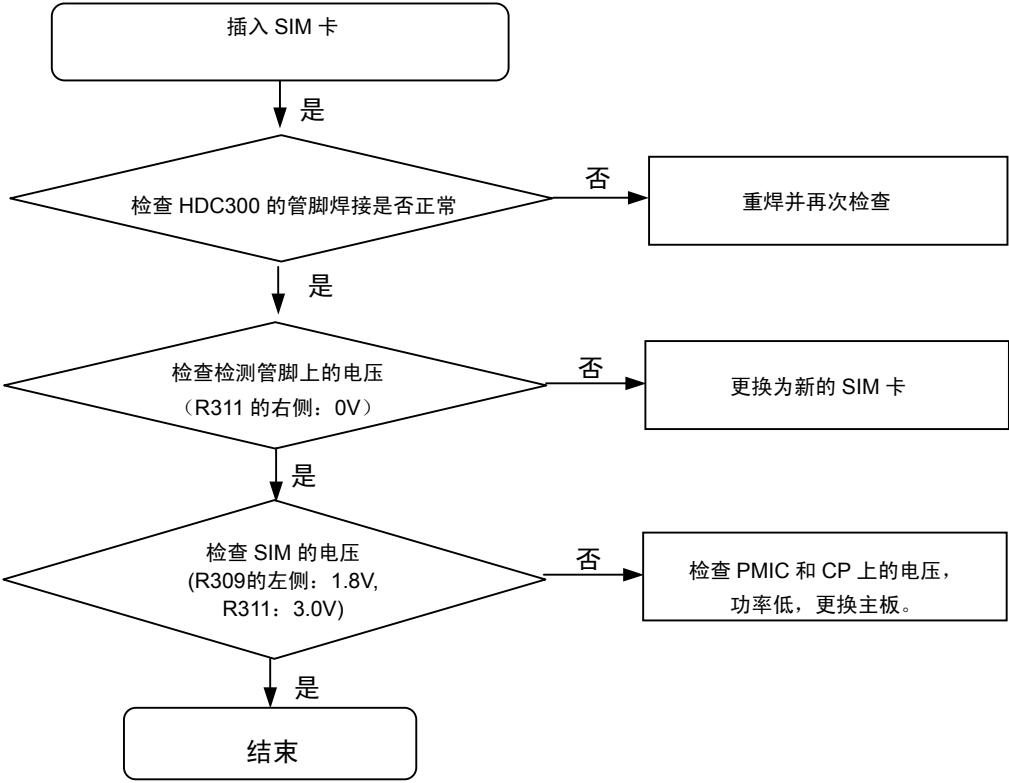


8-3-3. 充电部分

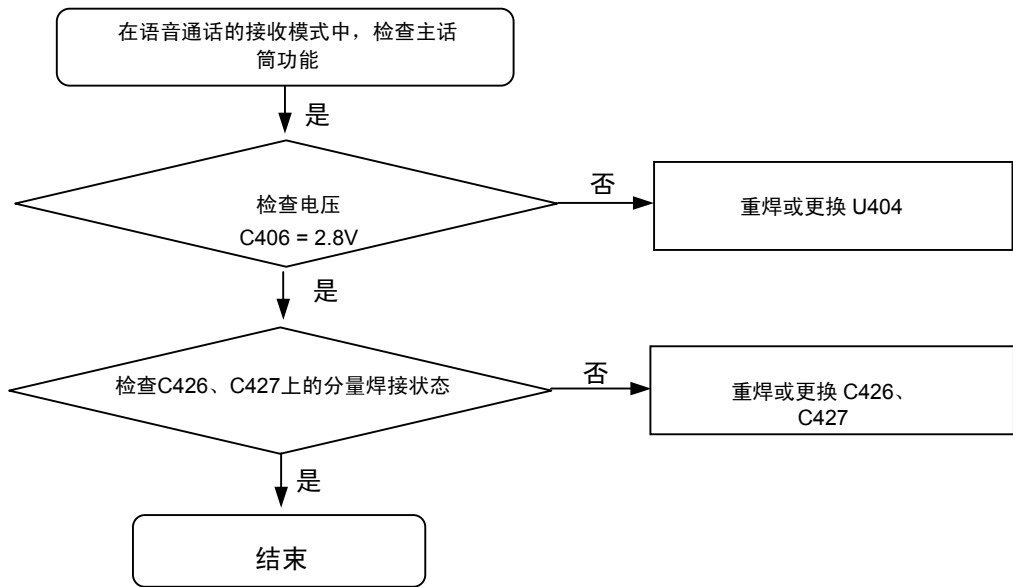


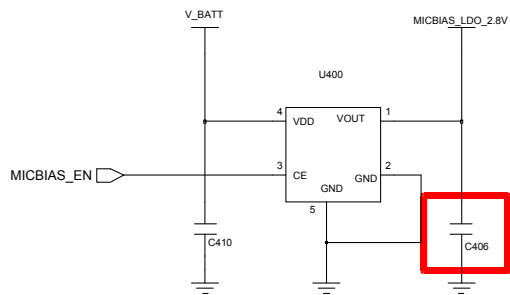
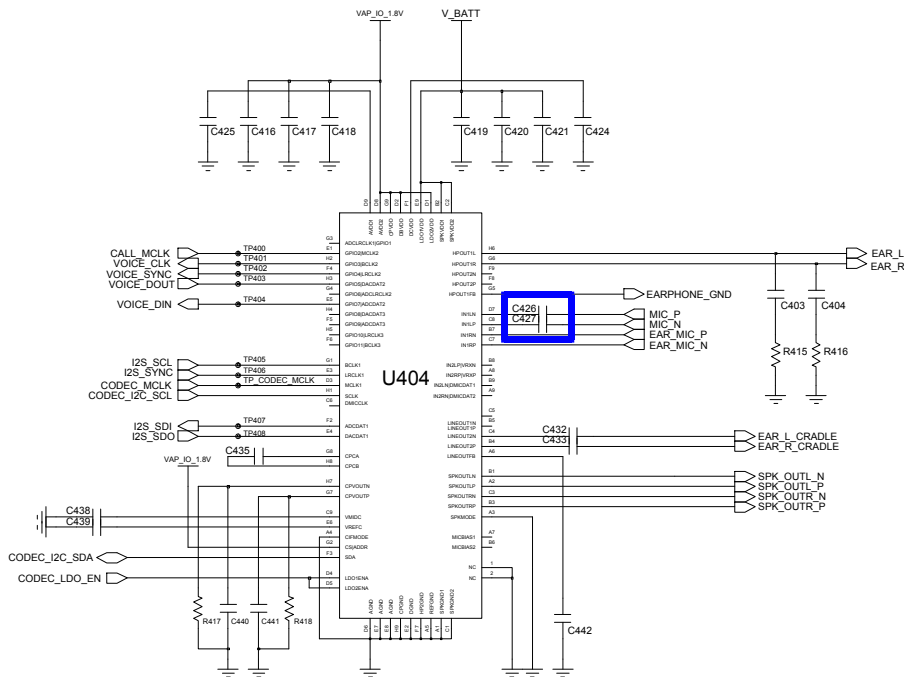


8-3-4.Sim 部分

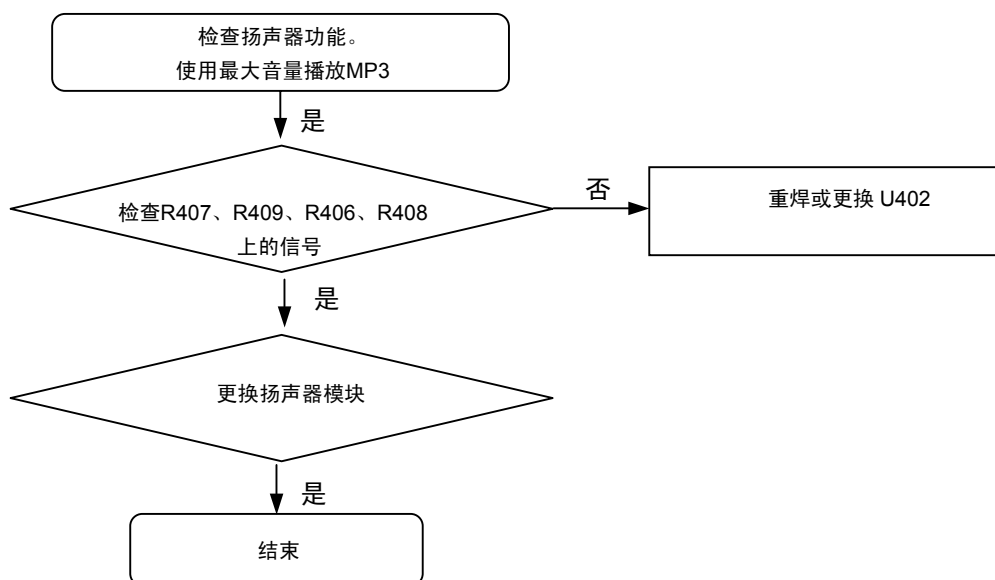


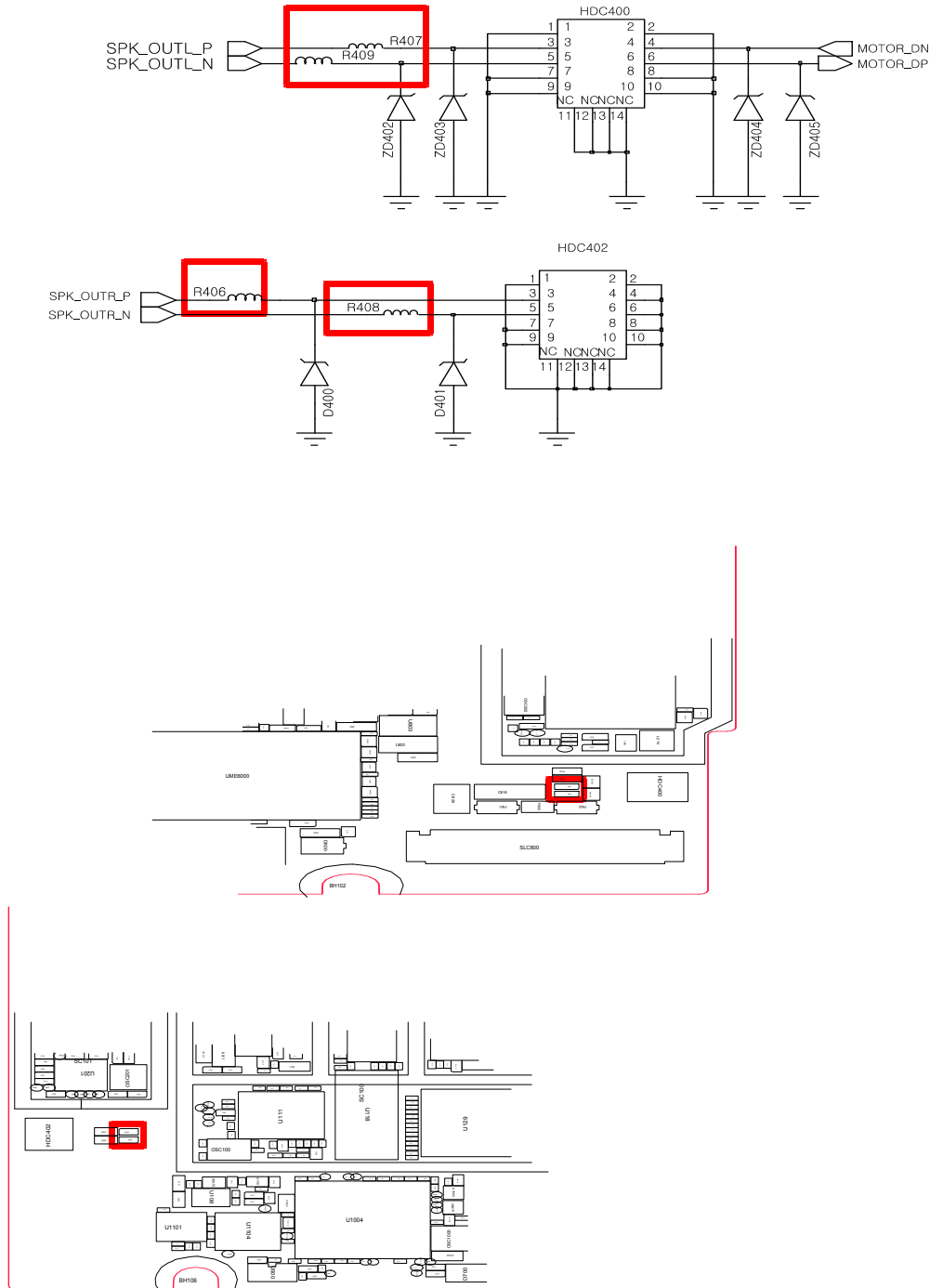
8-3-5. 话筒部分



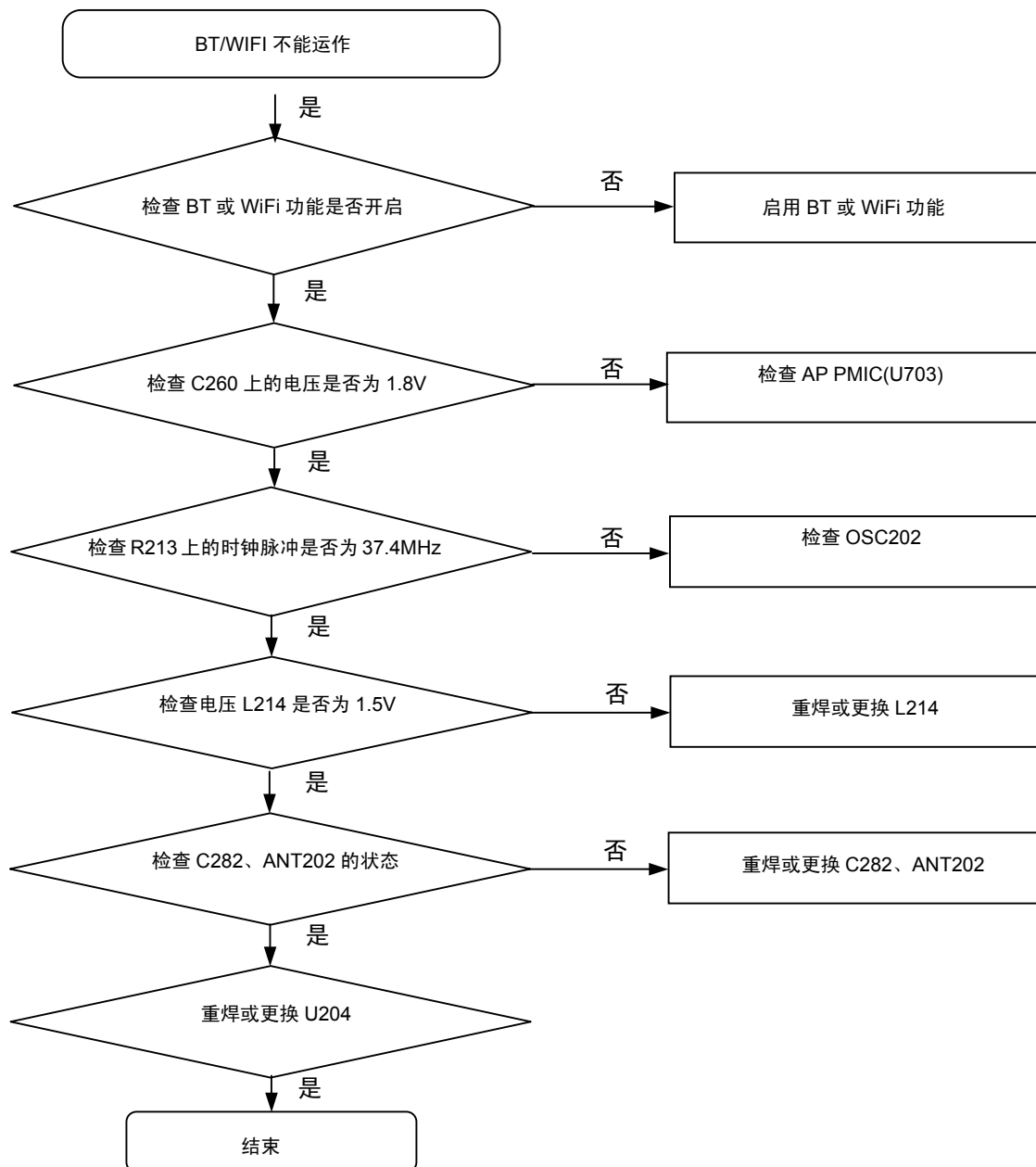


8-3-6.扬声器部分

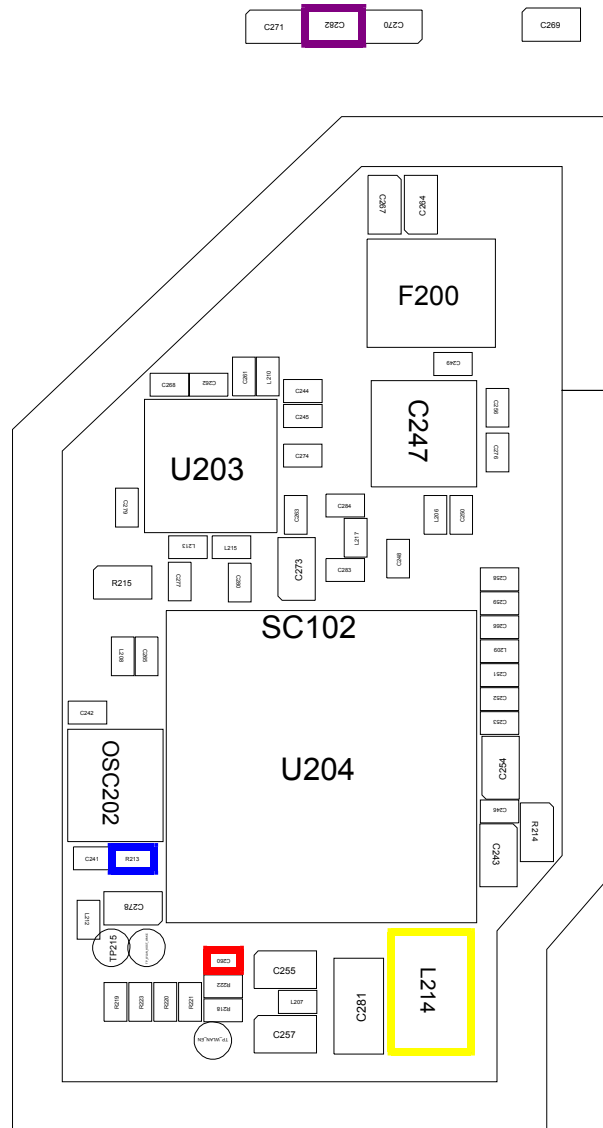




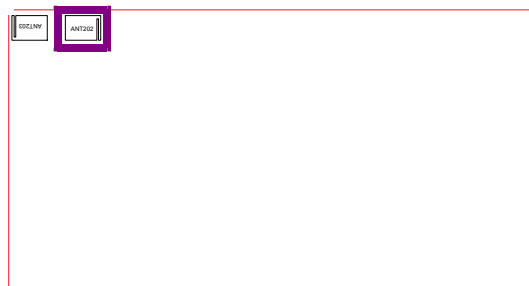
8-3-7 BT/WIFI





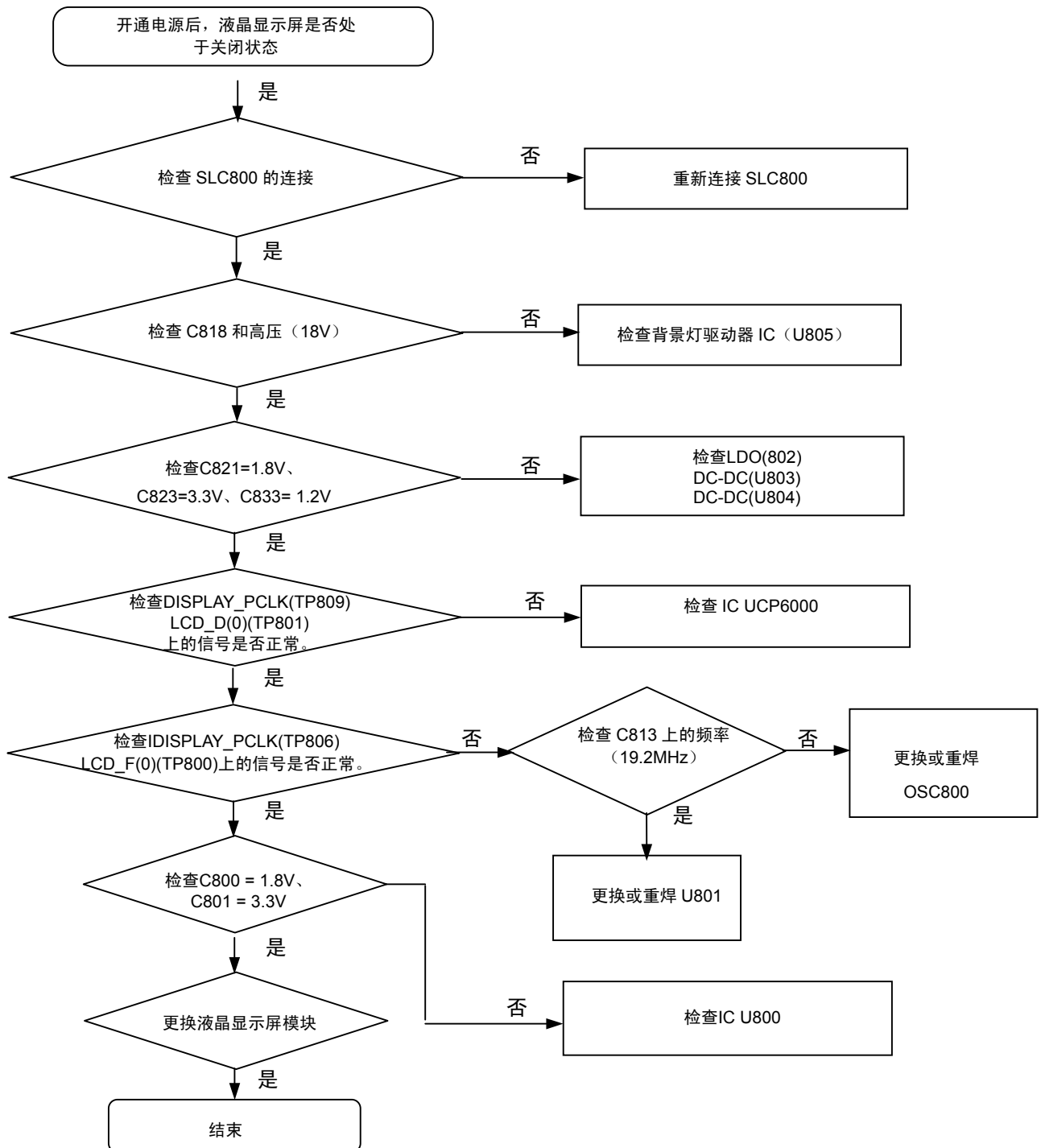


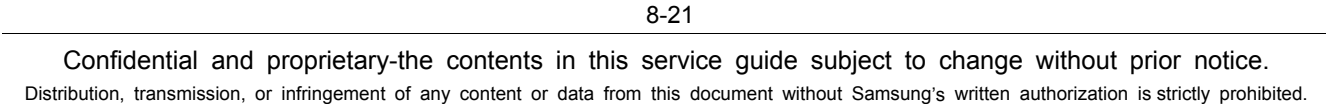
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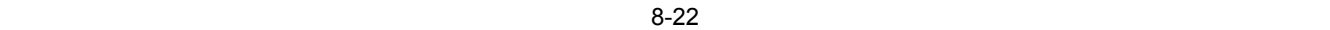


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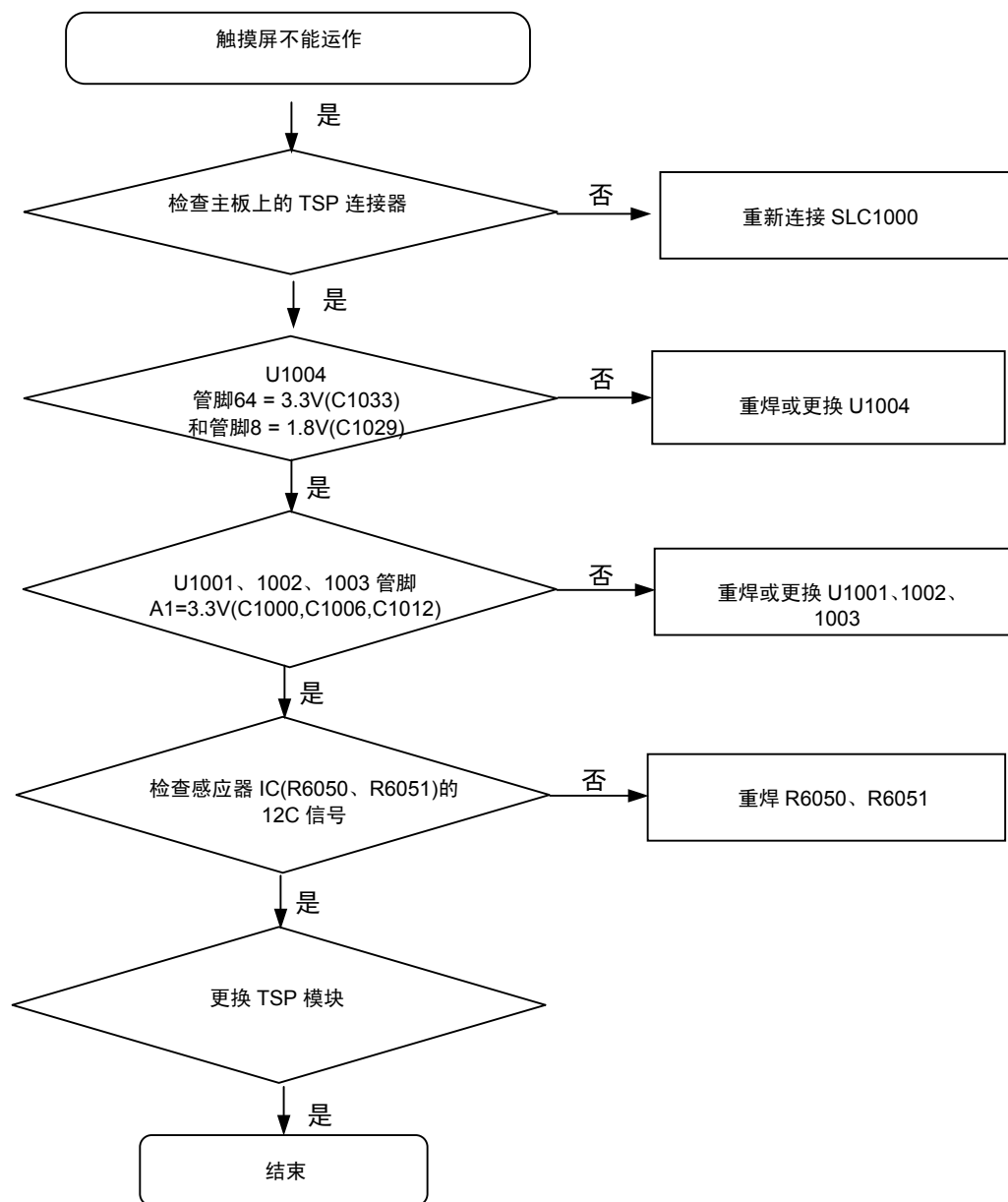
8-3-8.液晶显示屏



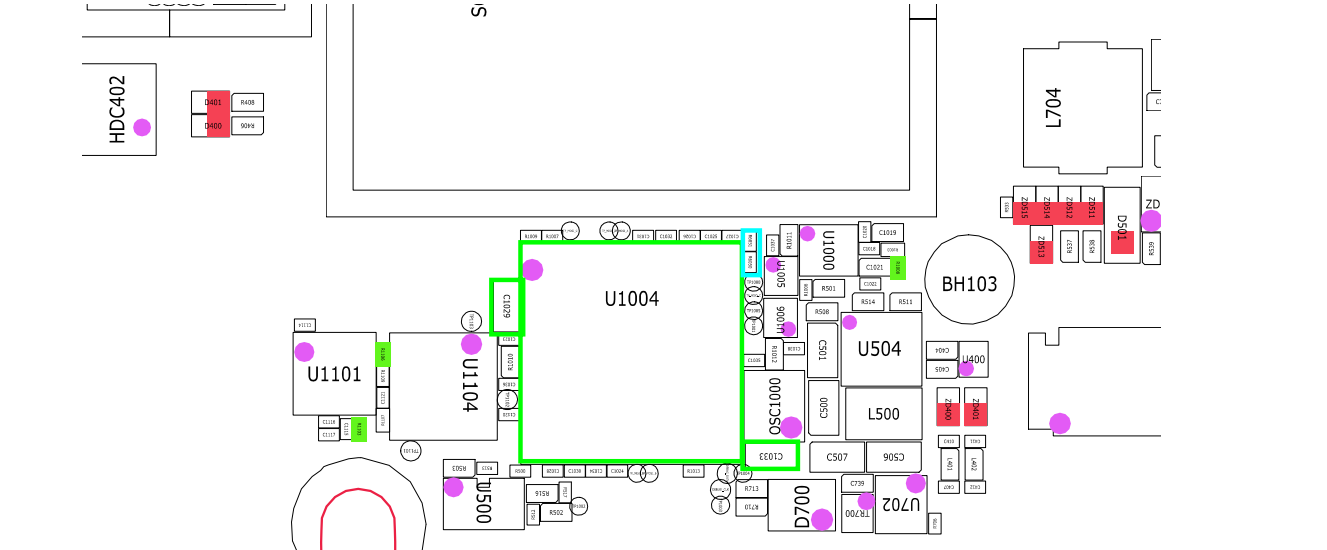




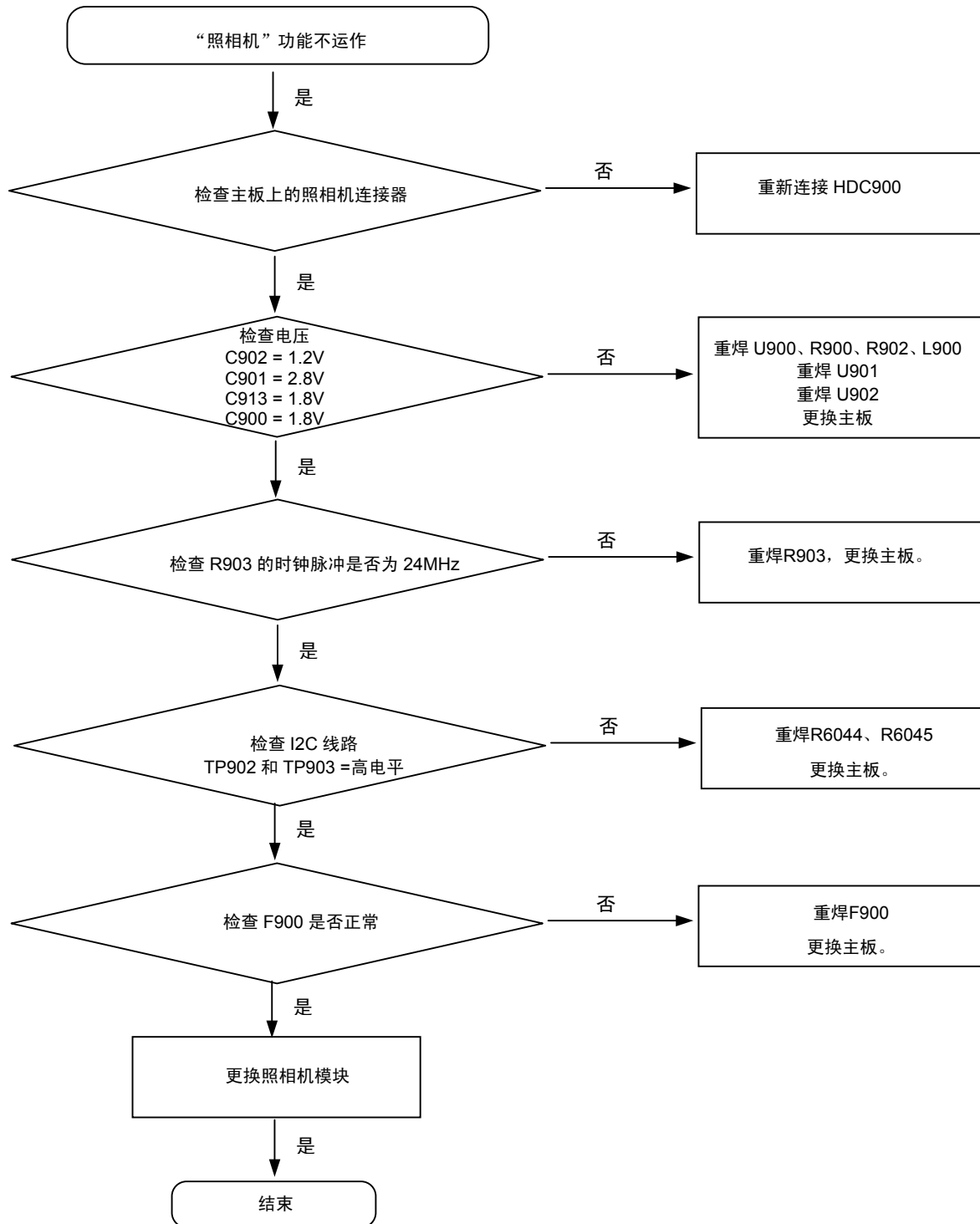
8-3-9.TSP

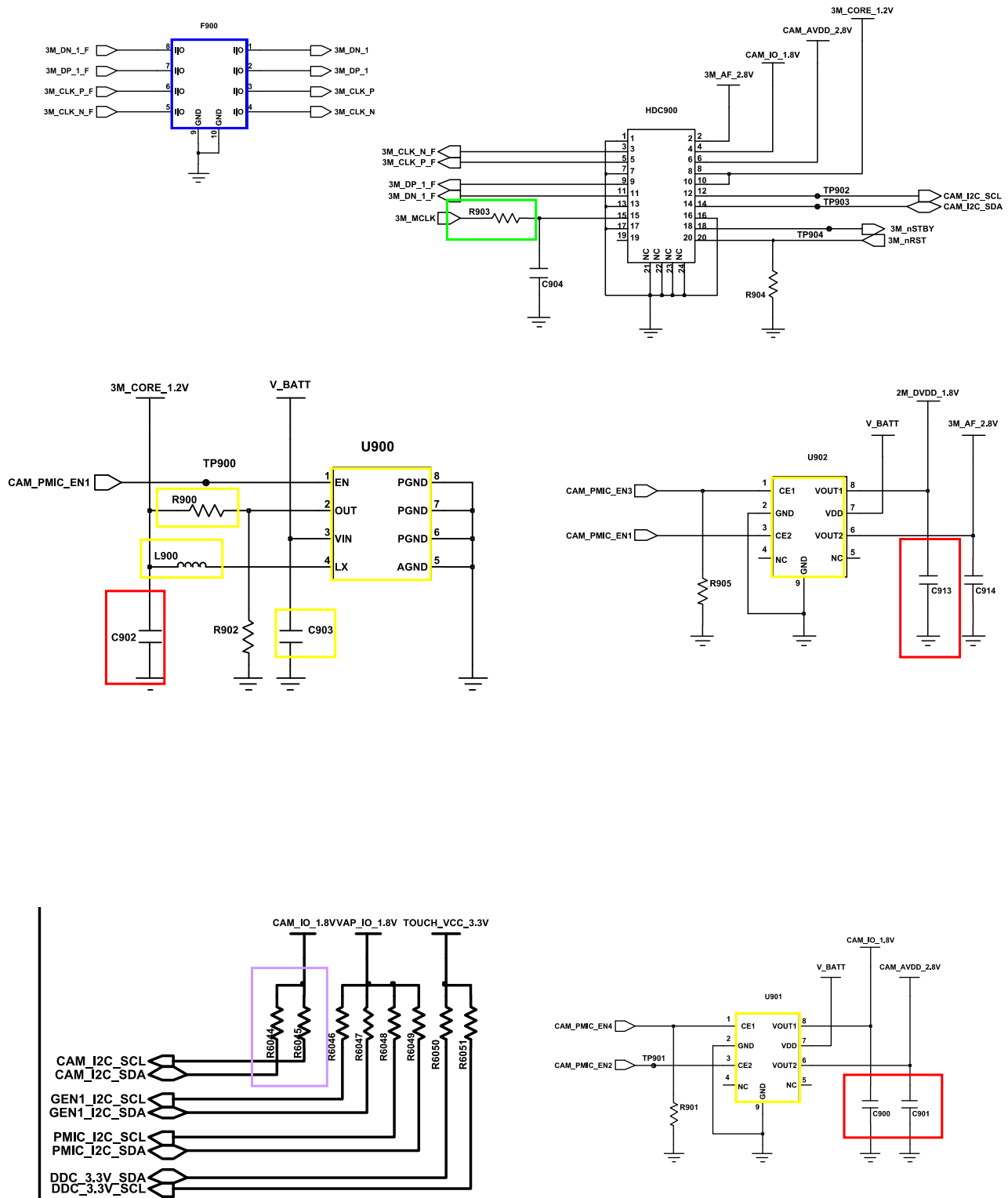






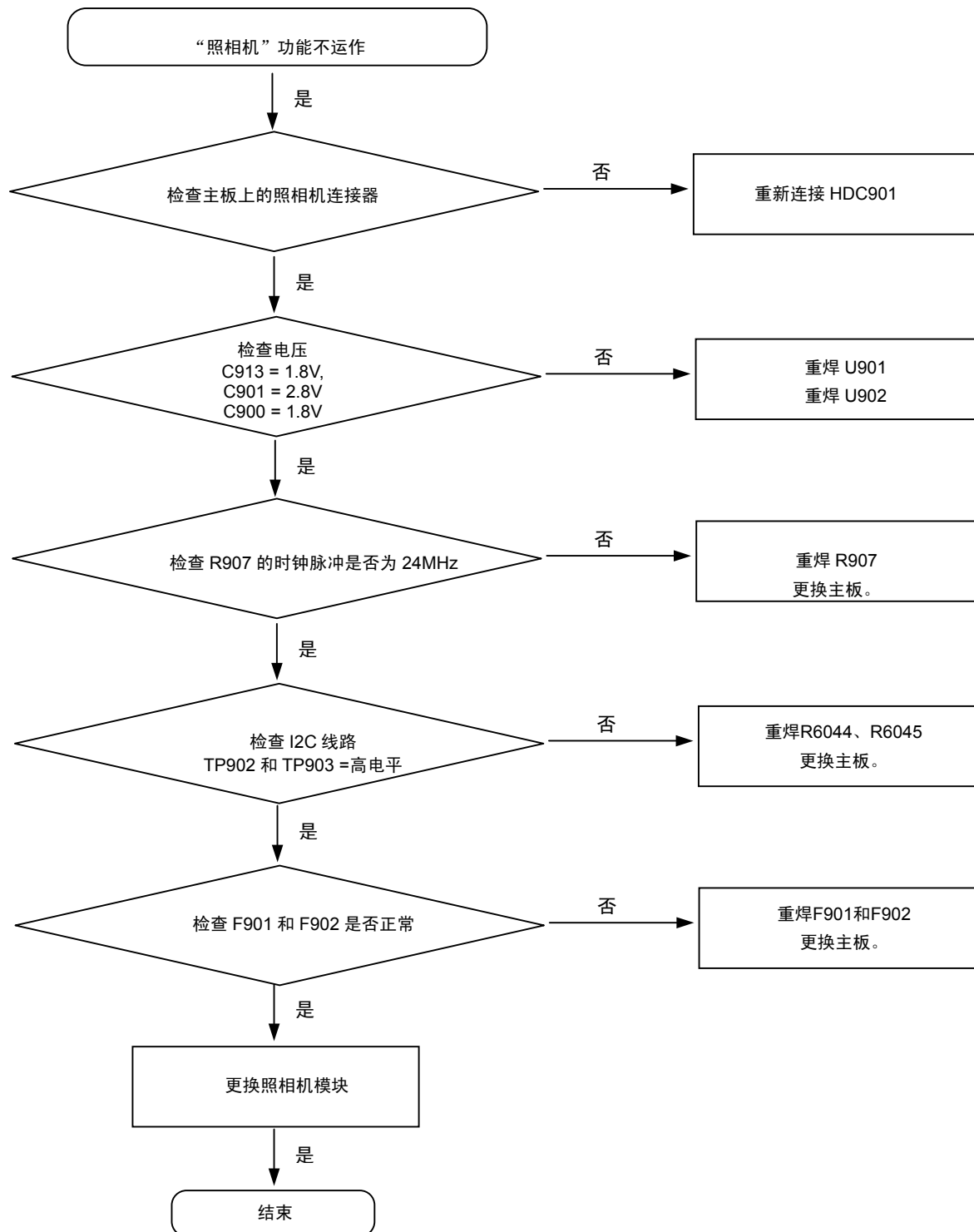
8-3-10.3M 照相机

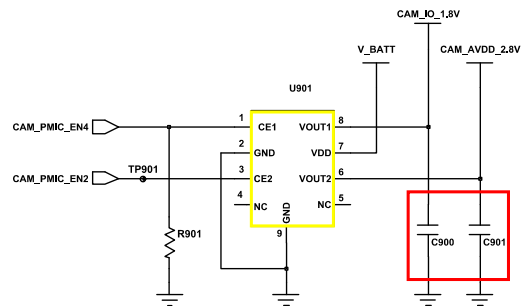
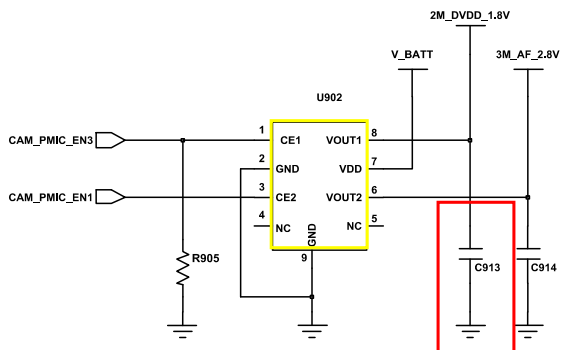
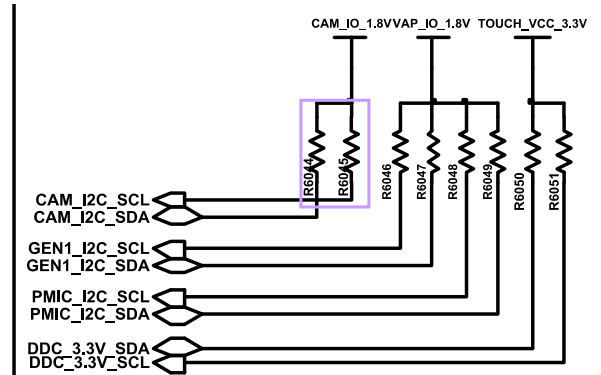
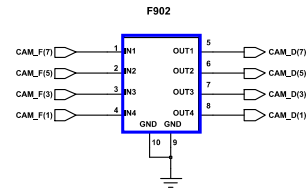
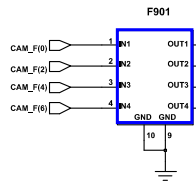
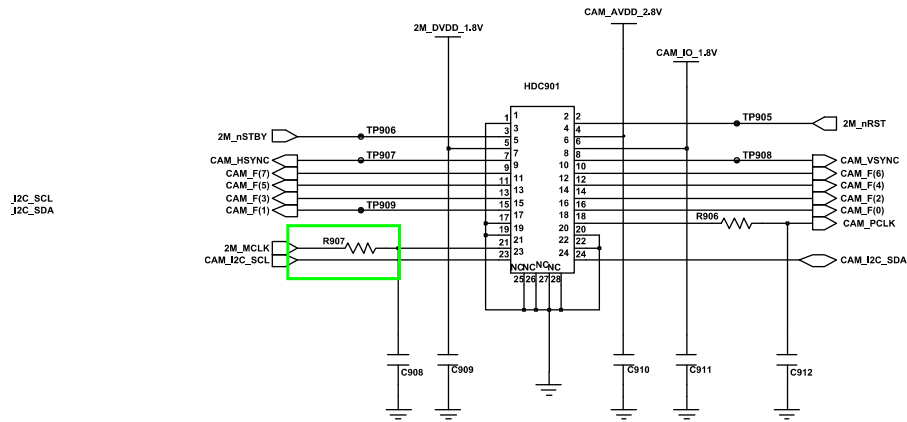


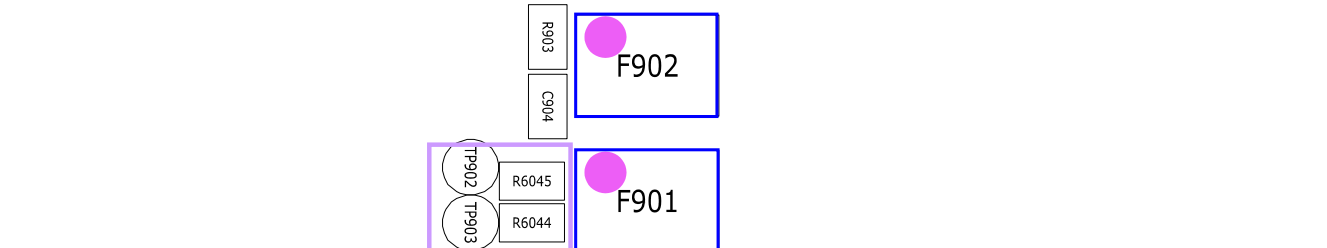




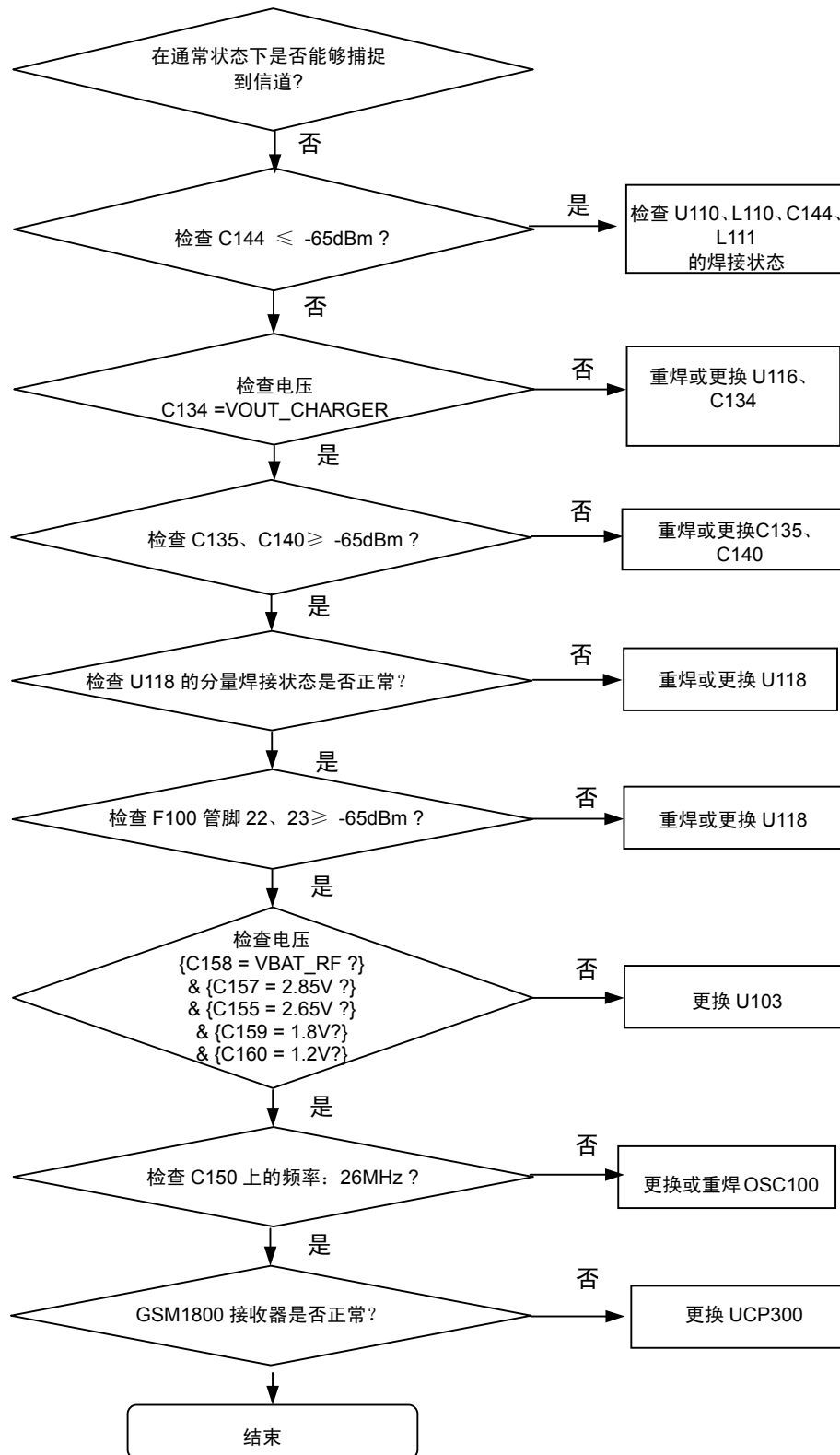
8-3-11. 2M 照相机



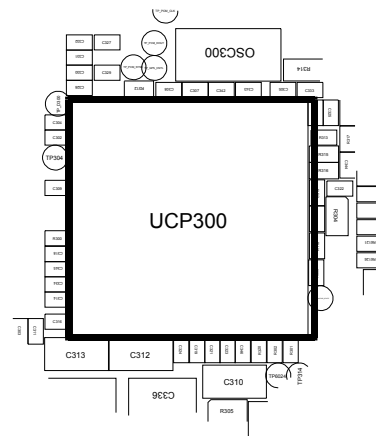
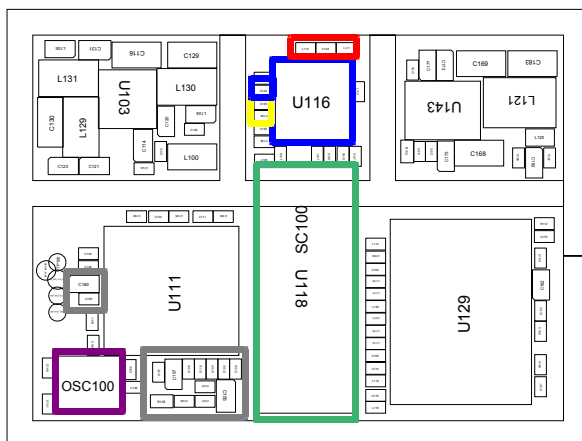
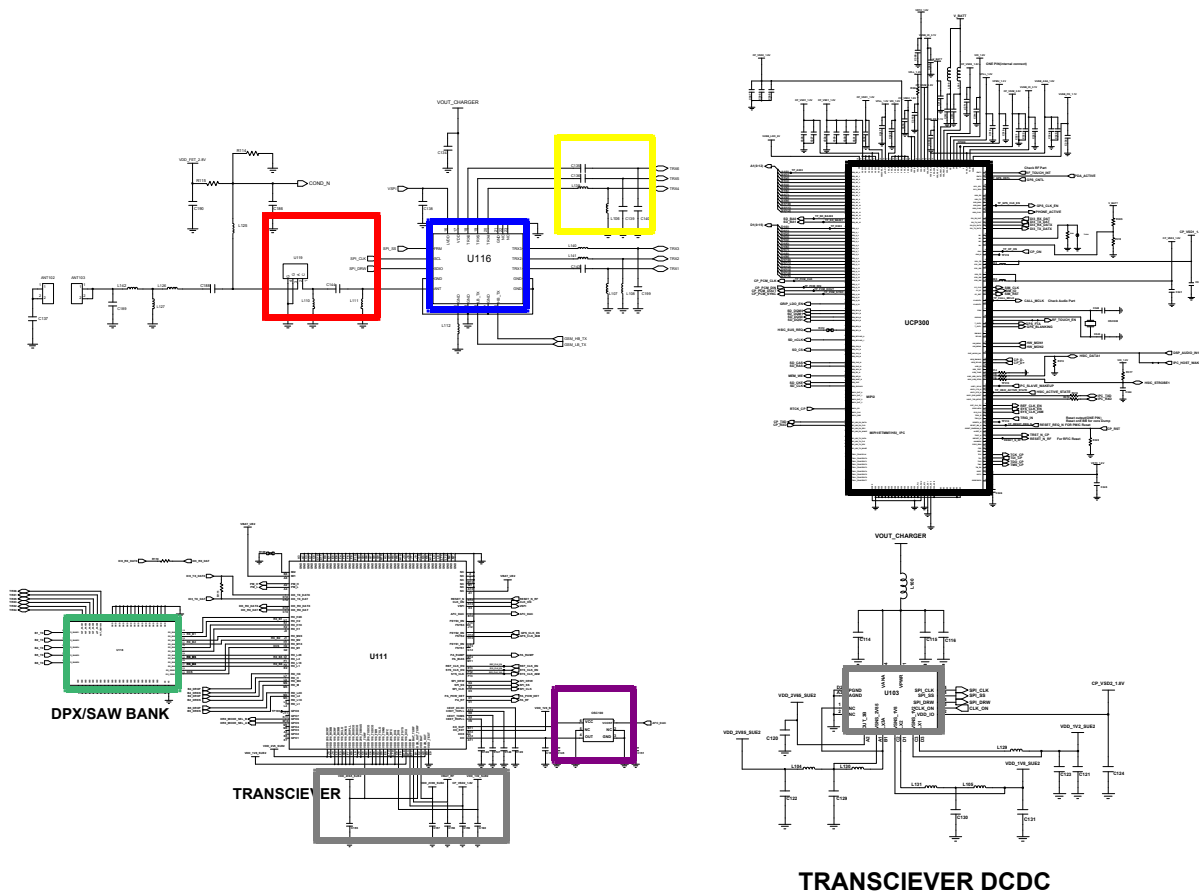




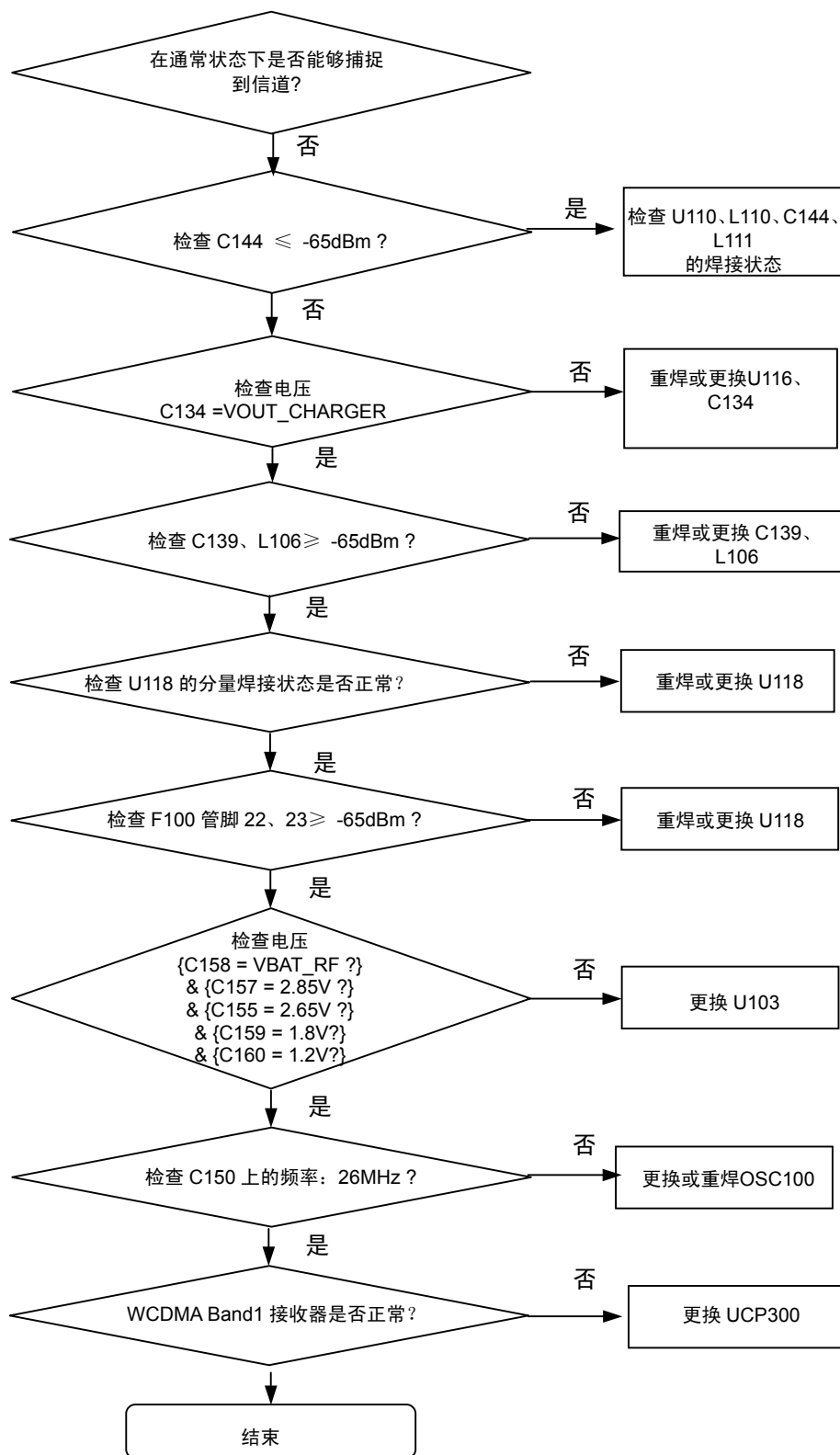
8-3-12.GSM1800 接收器



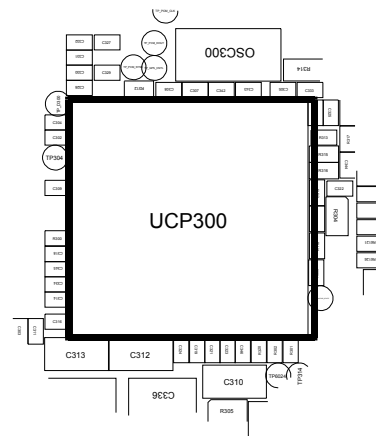
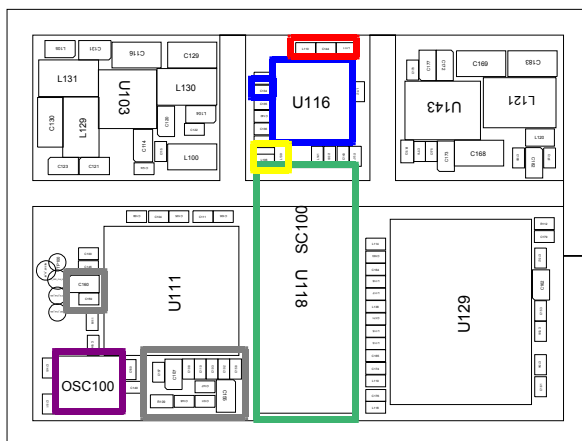
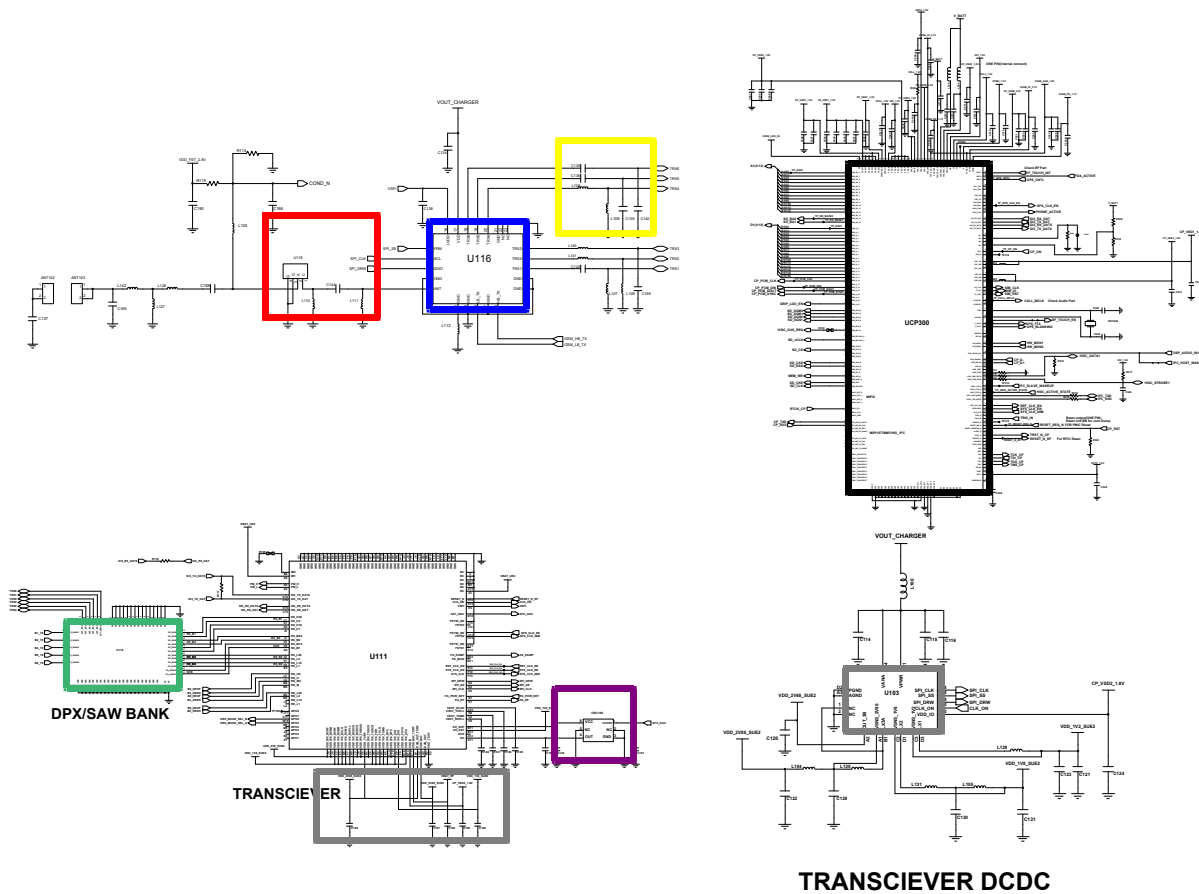
连续接收开通
射频输入: 698CH
功率: -50dBm



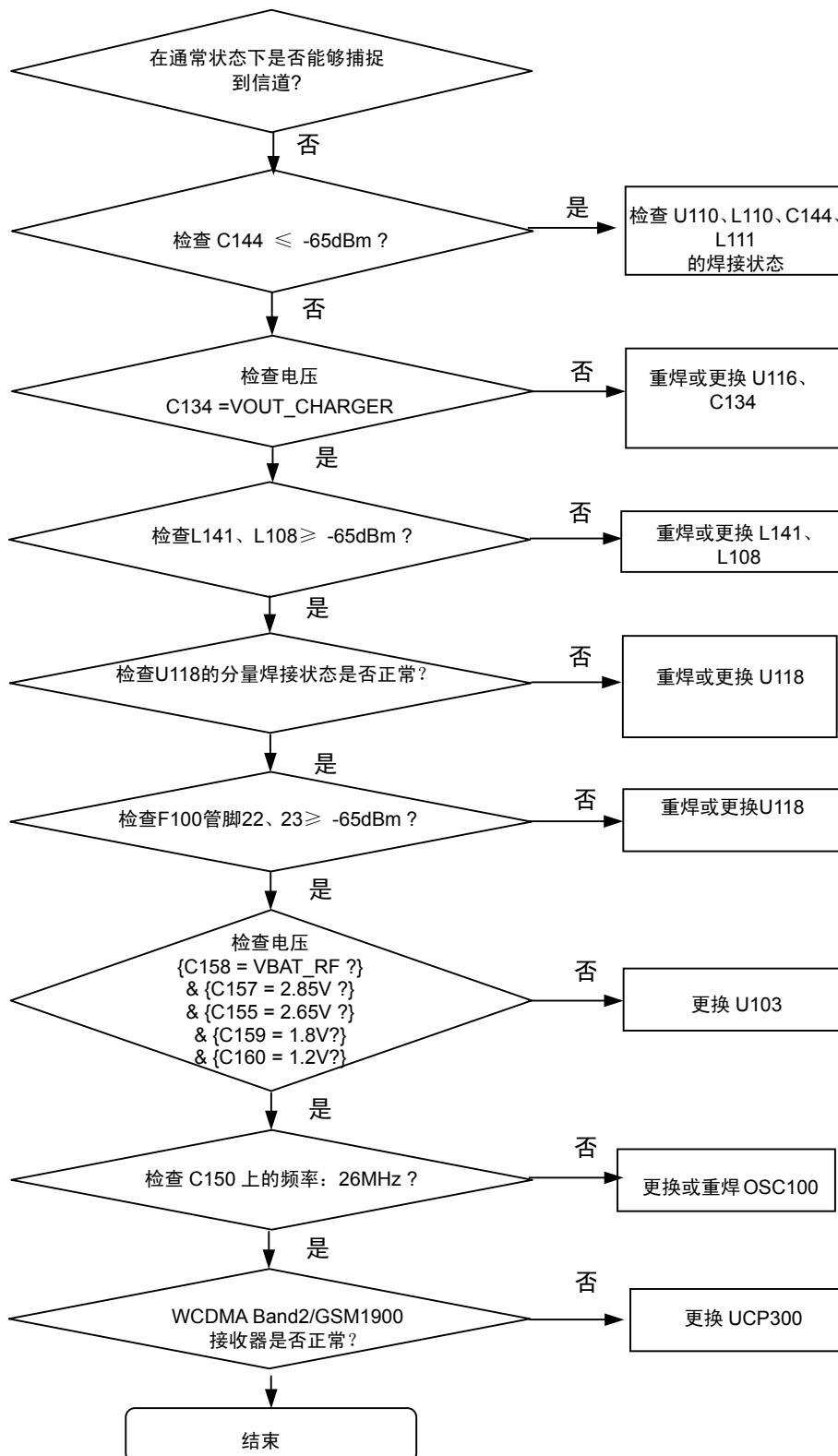
8-3-13.WCDMA Band1 接收器



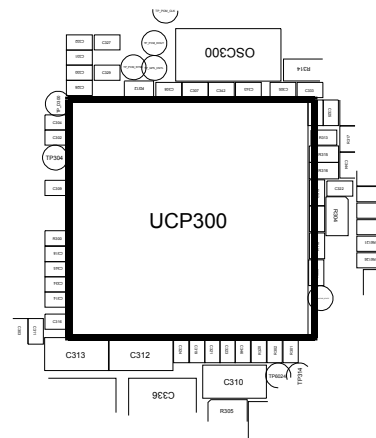
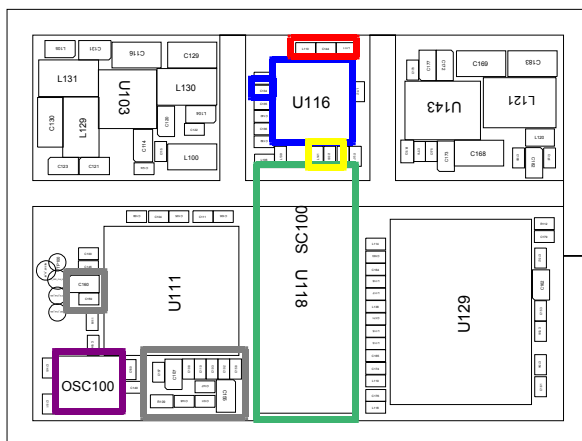
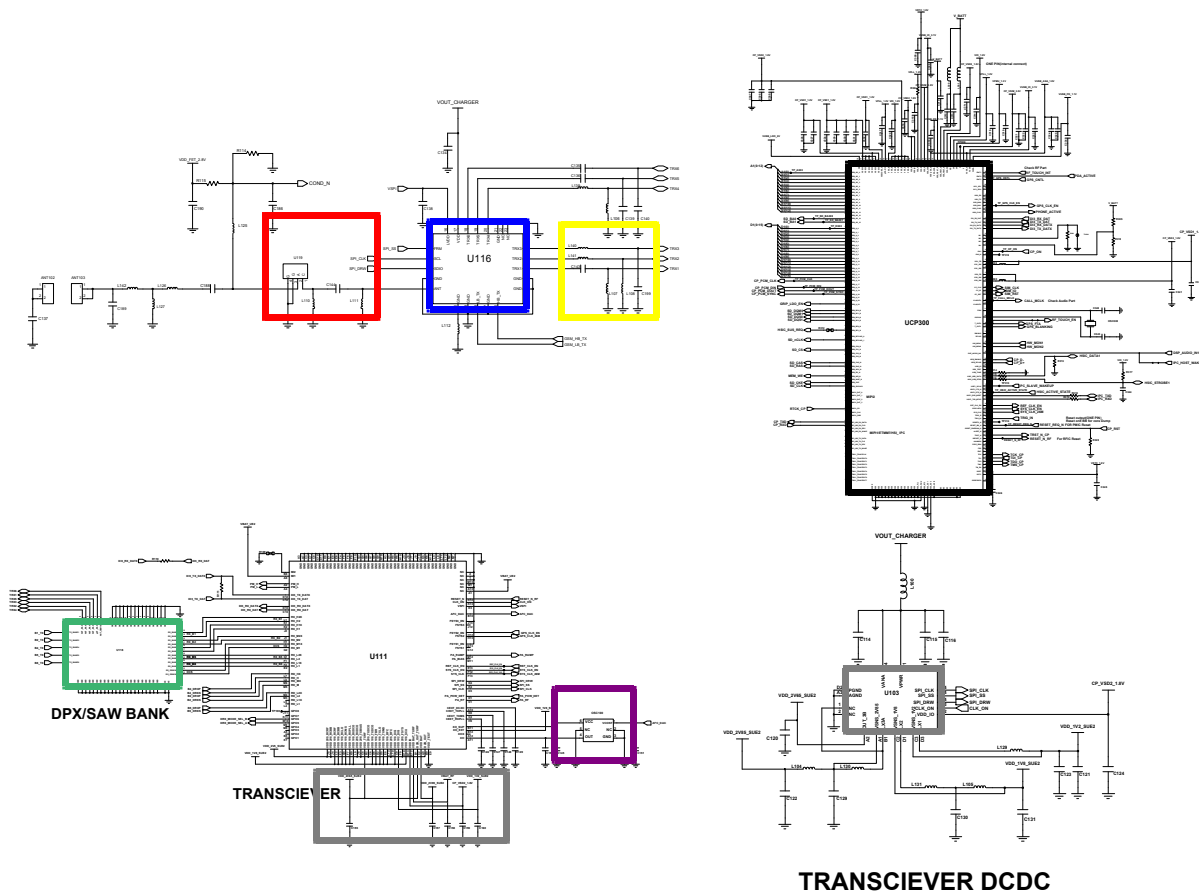
连续接收开通
射频输入: 10700CH
功率: -50dBm



8-3-14.WCDMA Band2/GSM1900 接收器

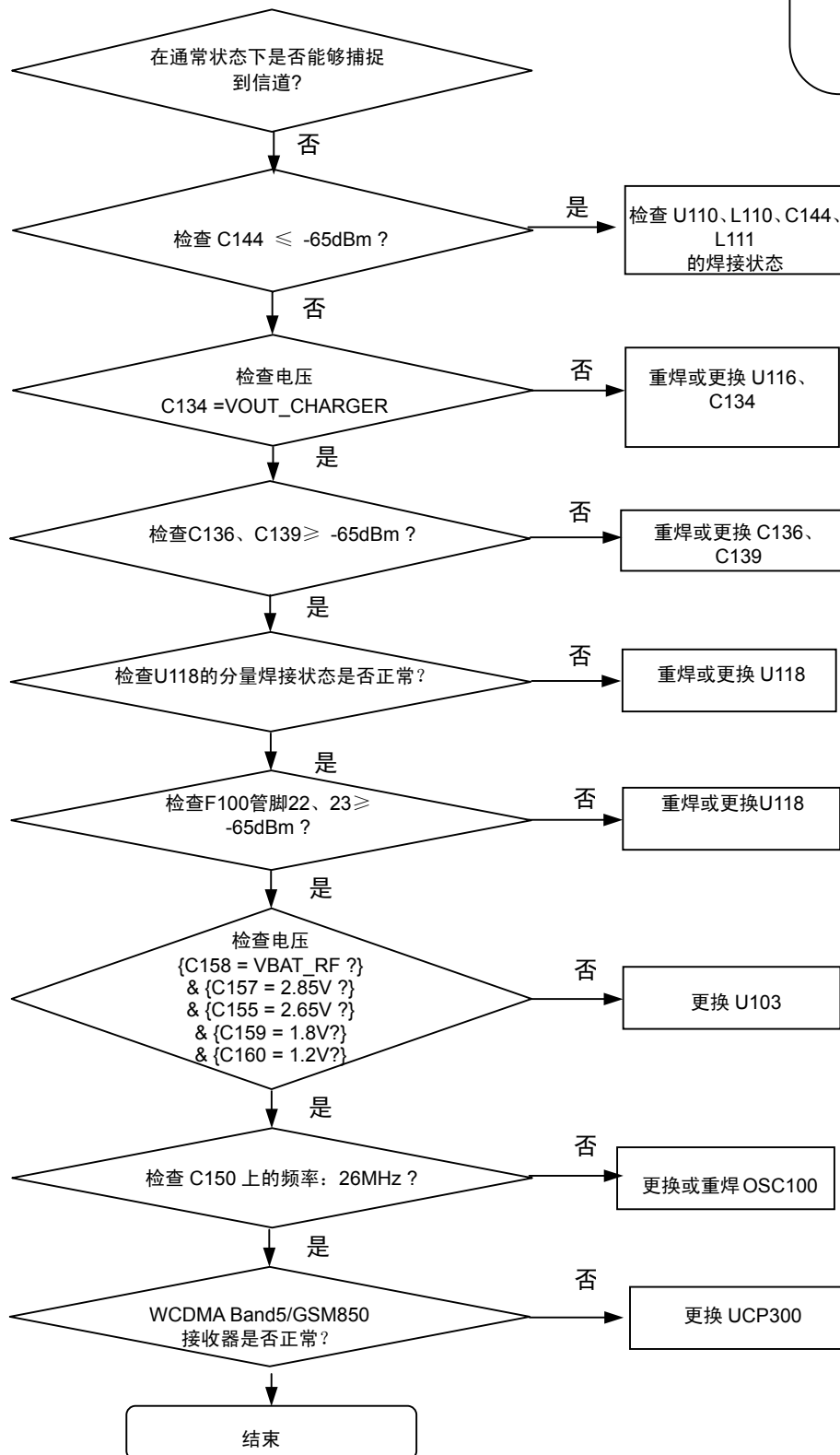


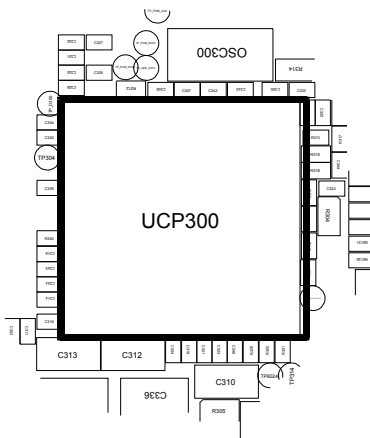
连续接收开通
射频输入: 9880CH
功率: -50dBm



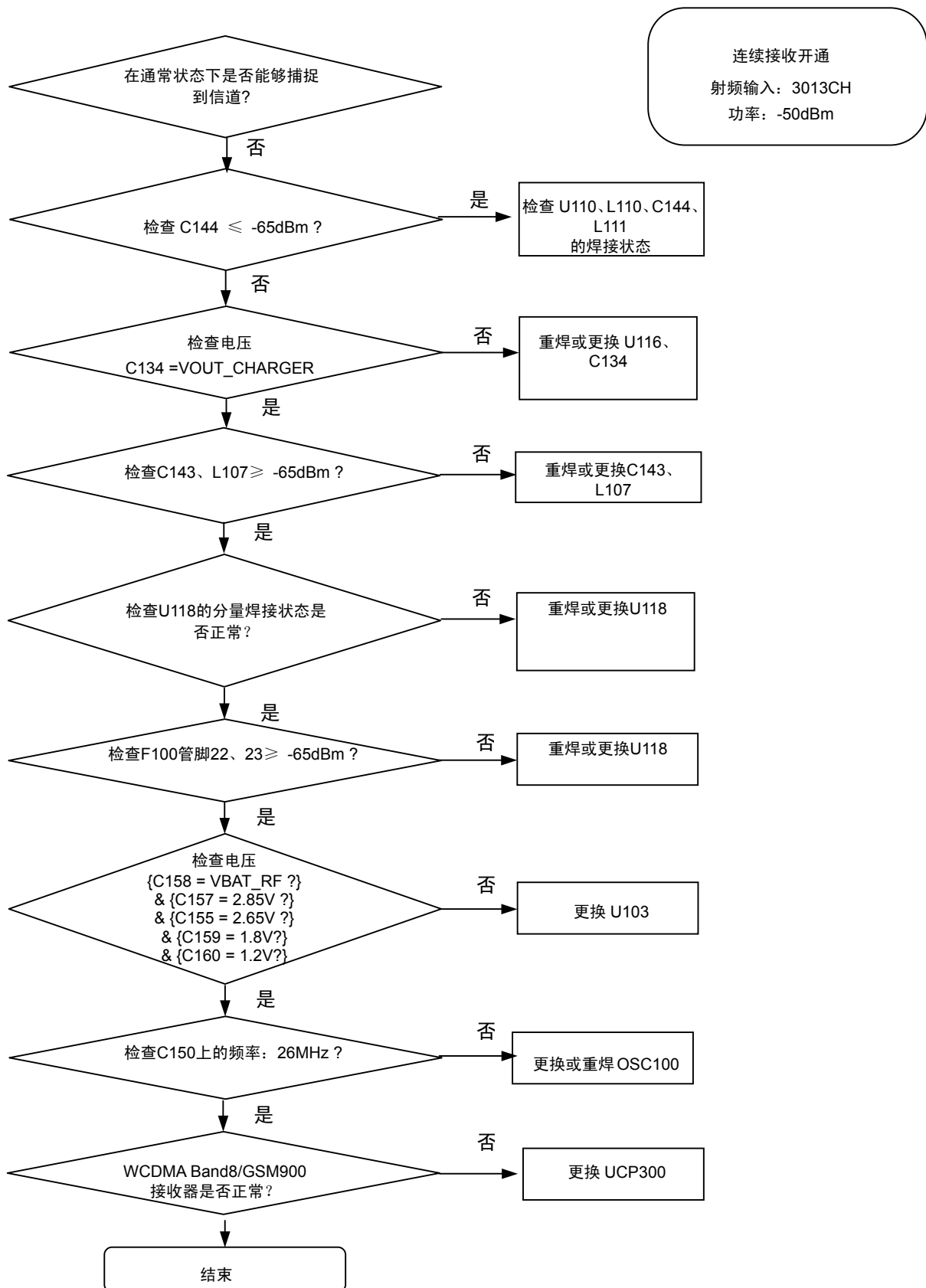
8-3-15.WCDMA Band5/GSM850 接收器

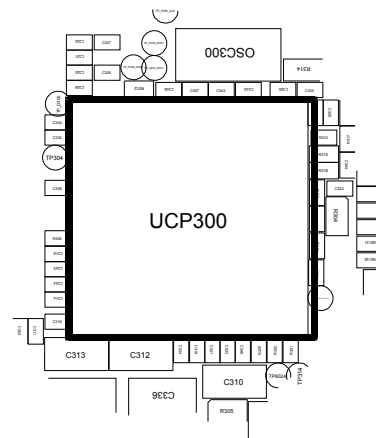
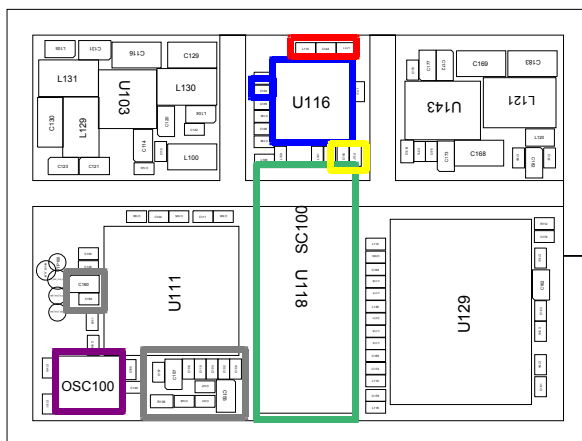
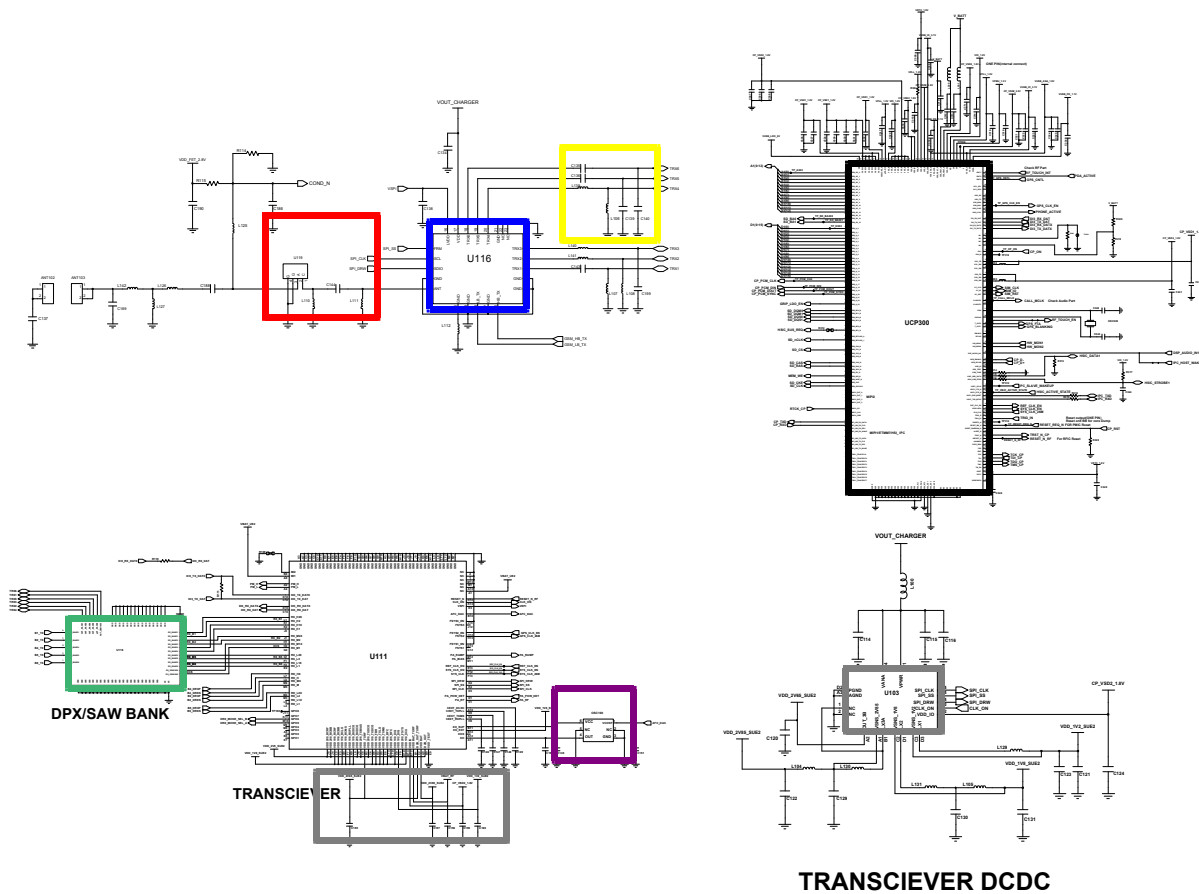
连续接收开通
射频输入: 4408CH
功率: -50dBm





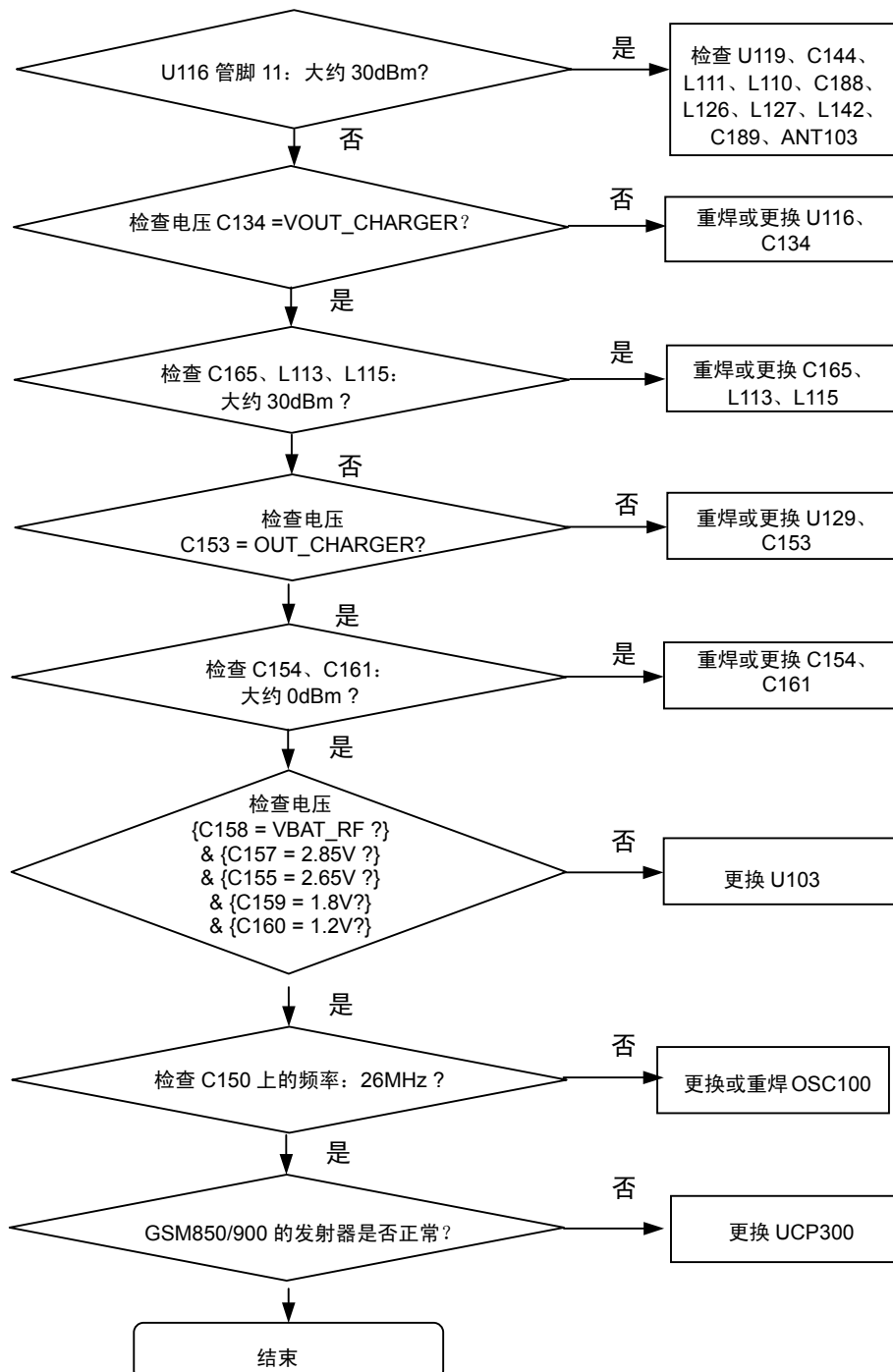
8-3-16.WCDMA Band8/GSM900 接收器

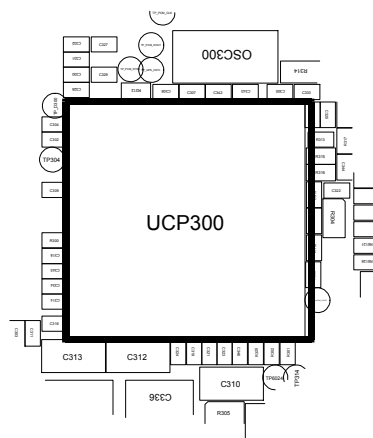
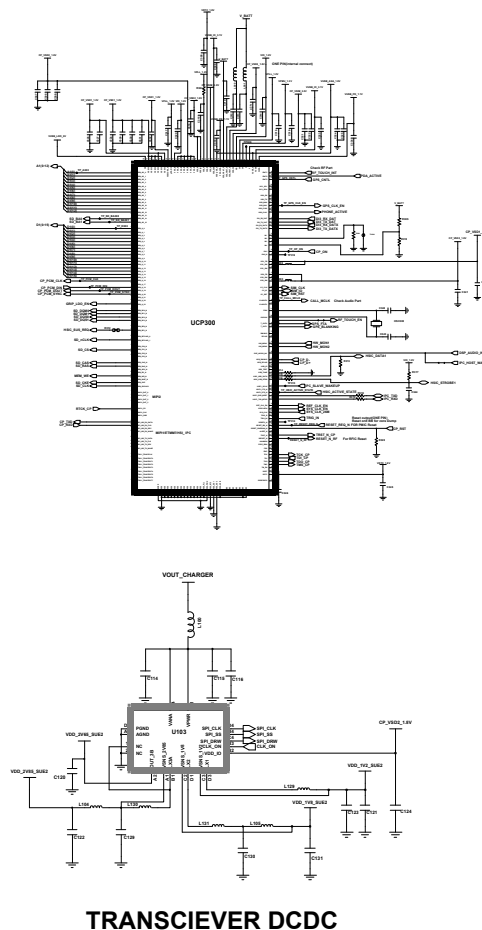




8-3-17.GSM850/GSM900 发射器

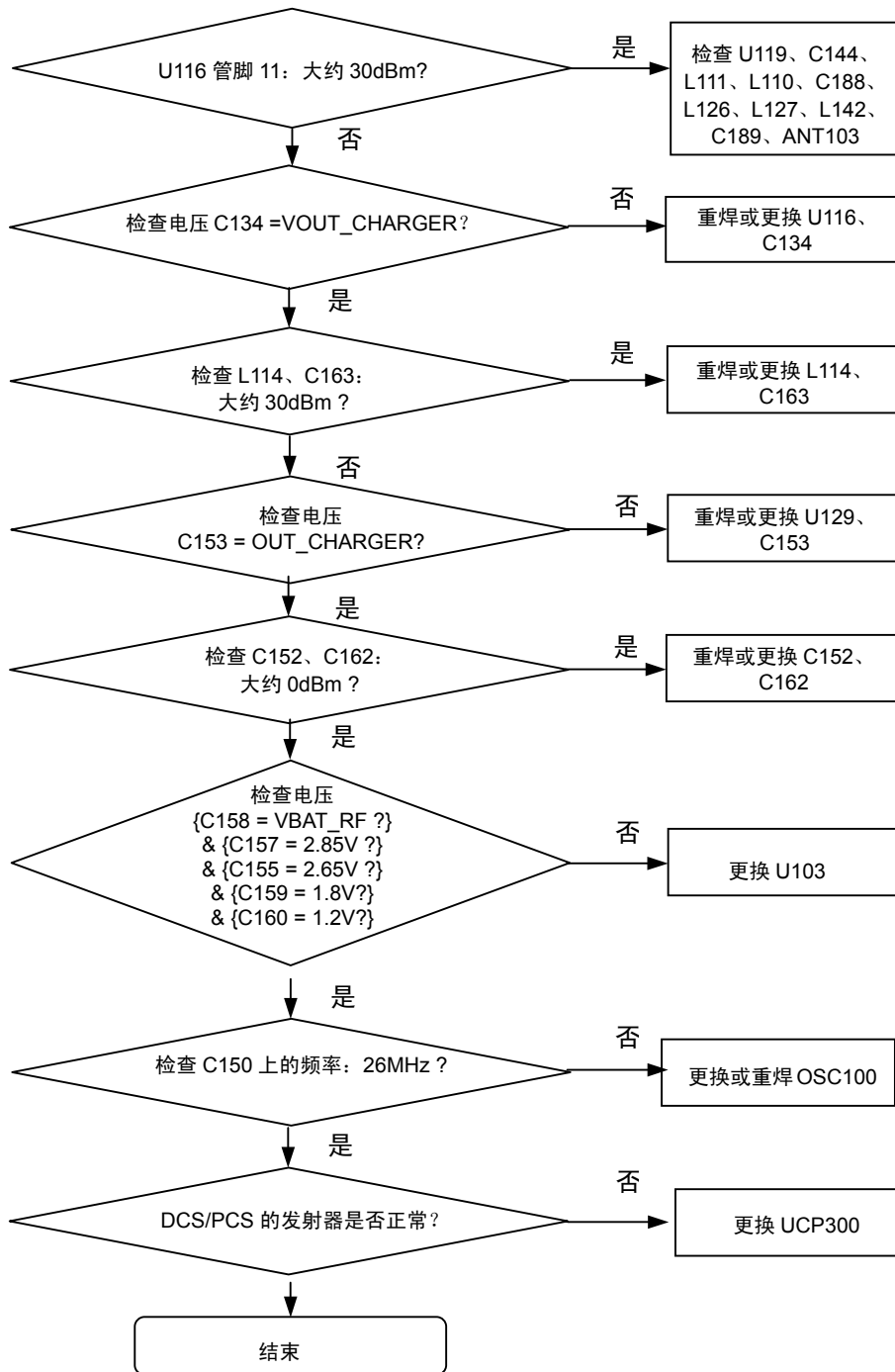
持续发射开通条件
 DAC 发射功率: 应用代码 14500
 GSM850 CH: 190
 GSM900 CH: 62
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB

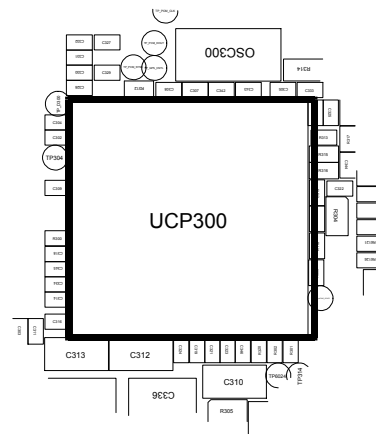
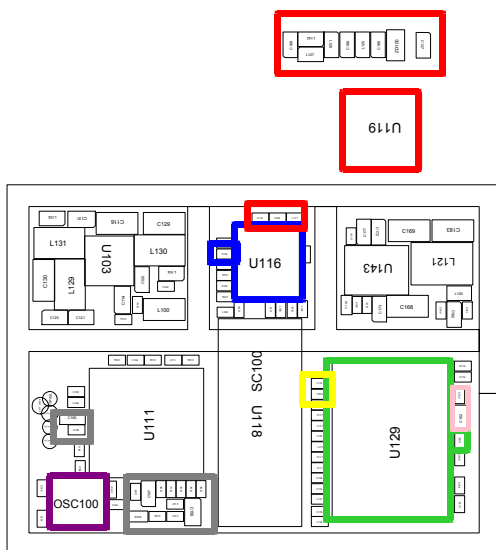
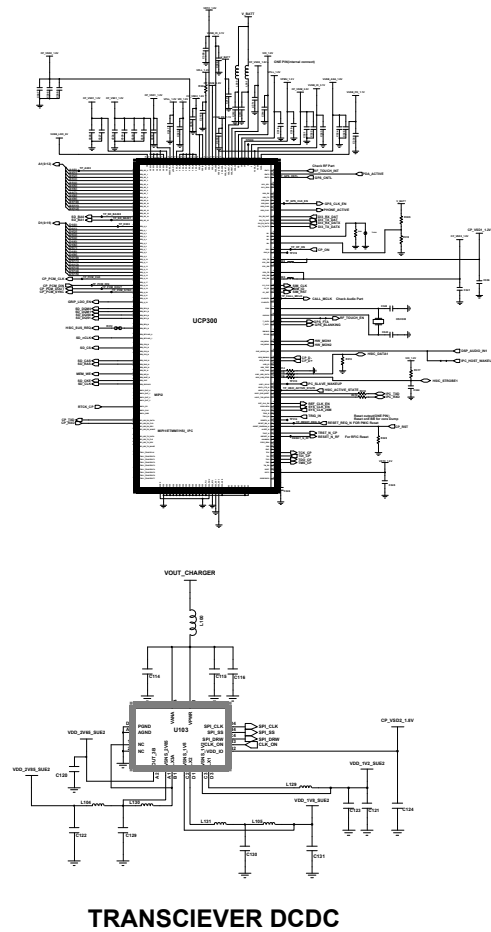
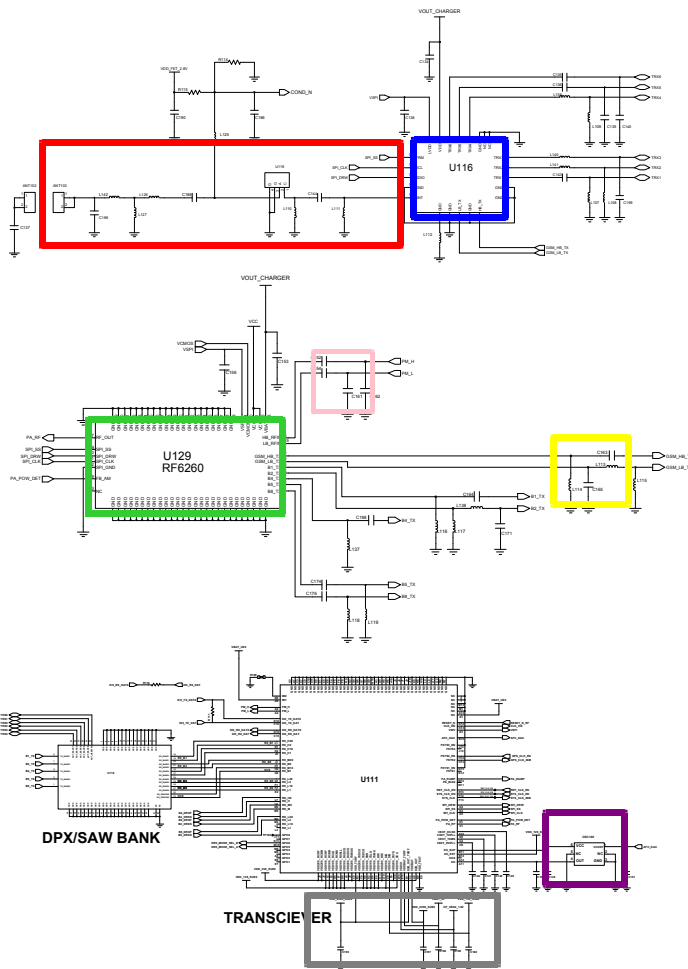




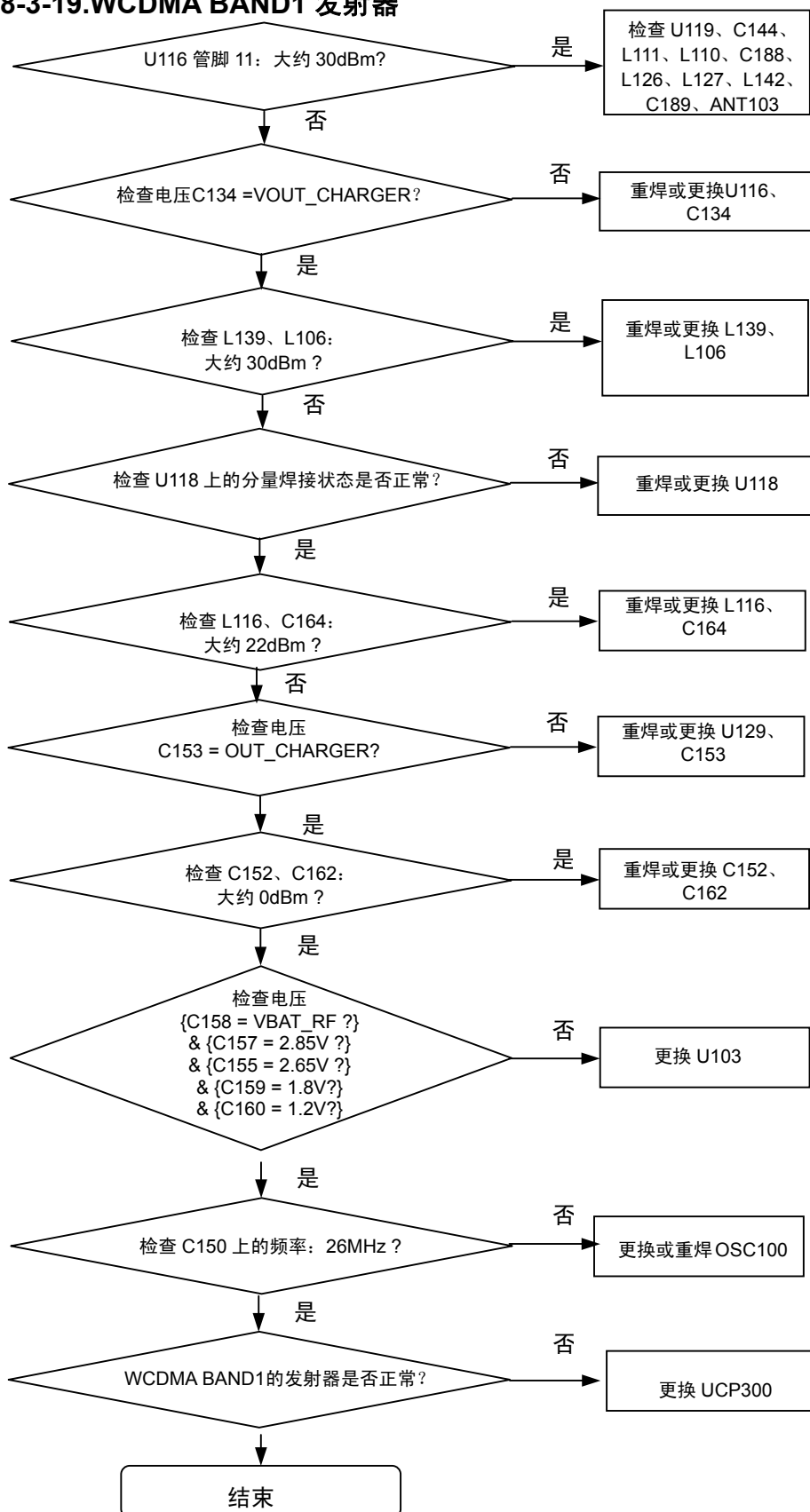
8-3-18.DCS/PCS 发射器

持续发射开通条件
 DAC 发射功率: 应用代码 14500
 DCS CH: 685
 PCS CH: 661
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB

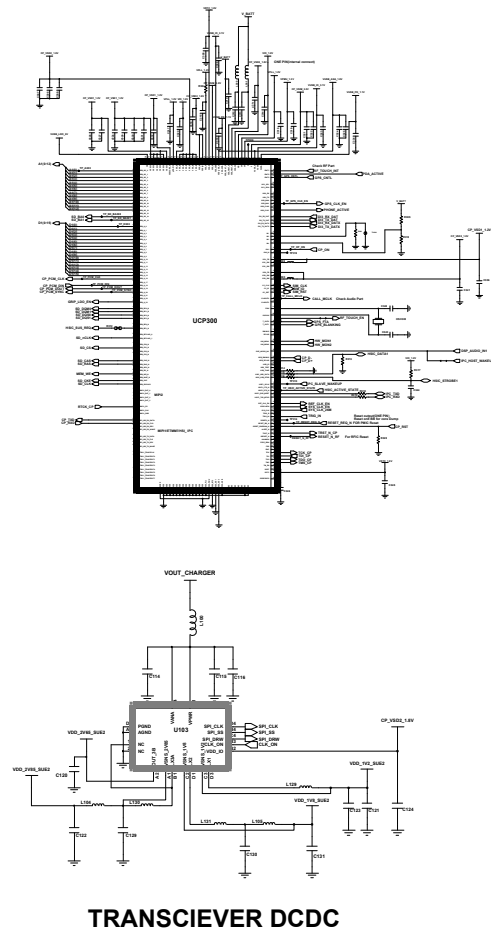
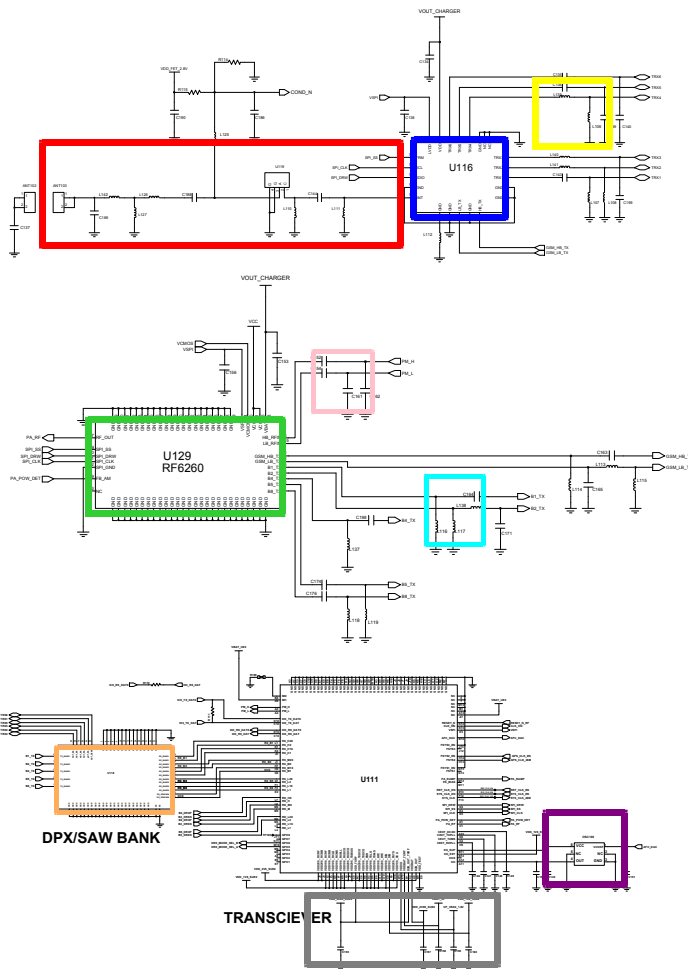




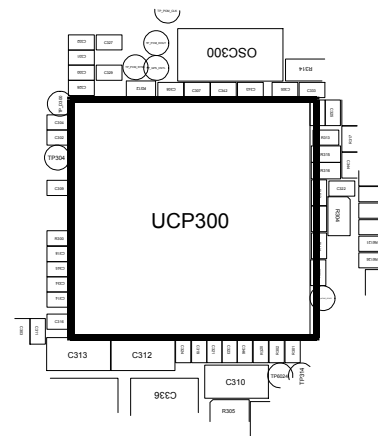
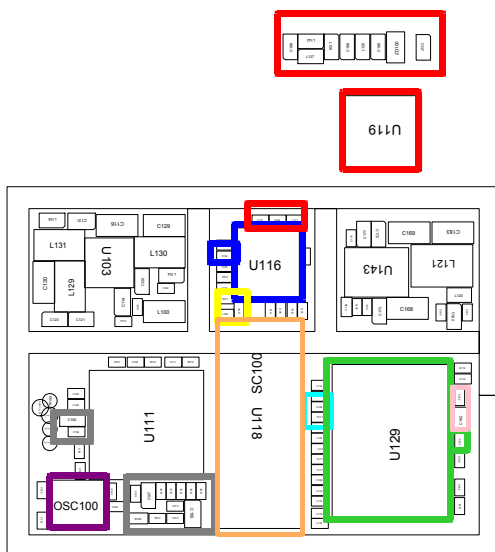
8-3-19.WCDMA BAND1 发射器



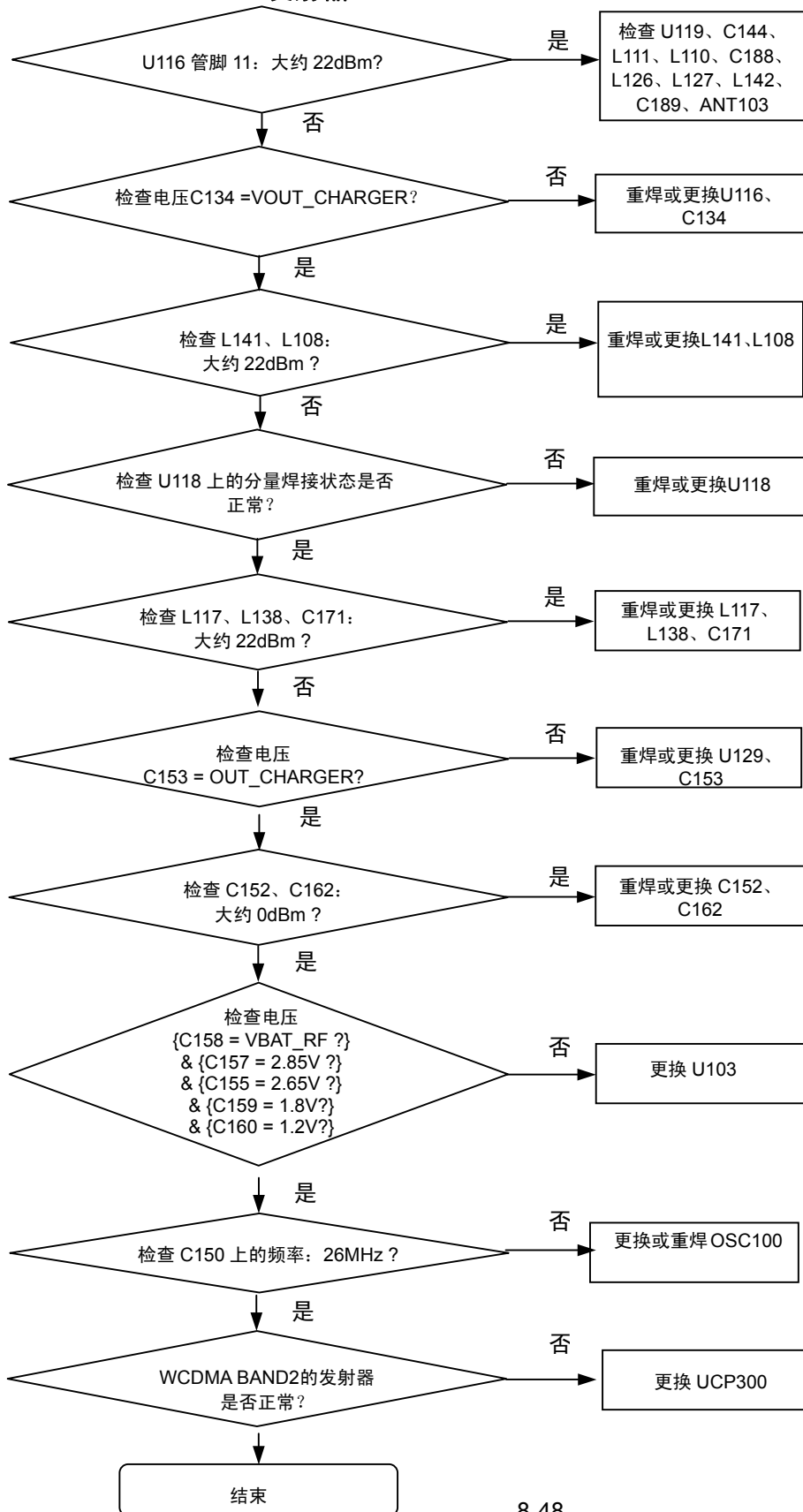
持续发射开通条件
 DAC 发射功率: 应用代码 14500
 WCDMA Band1 CH: 10700
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



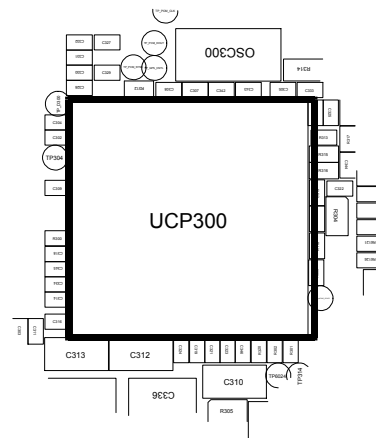
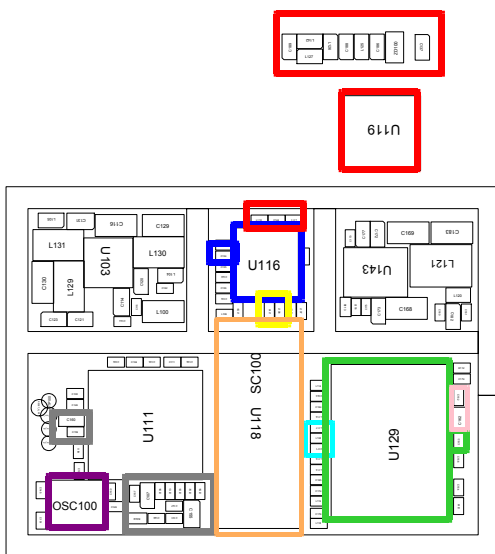
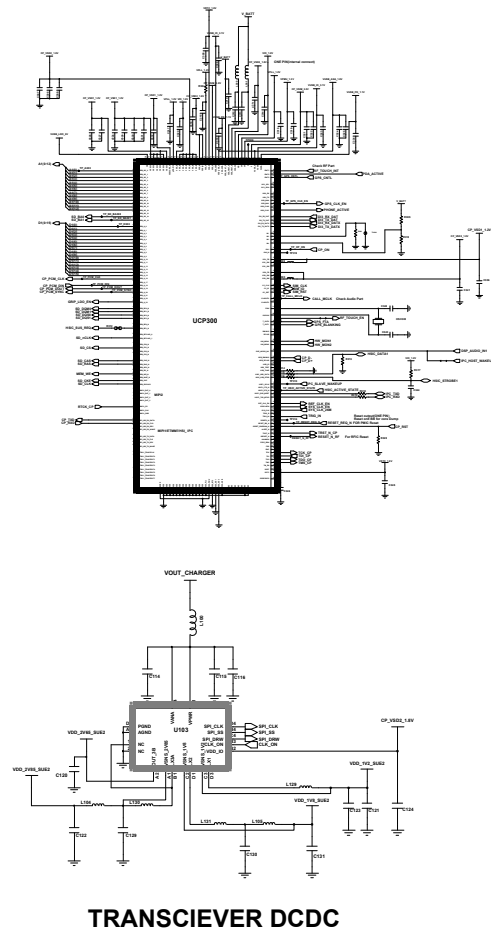
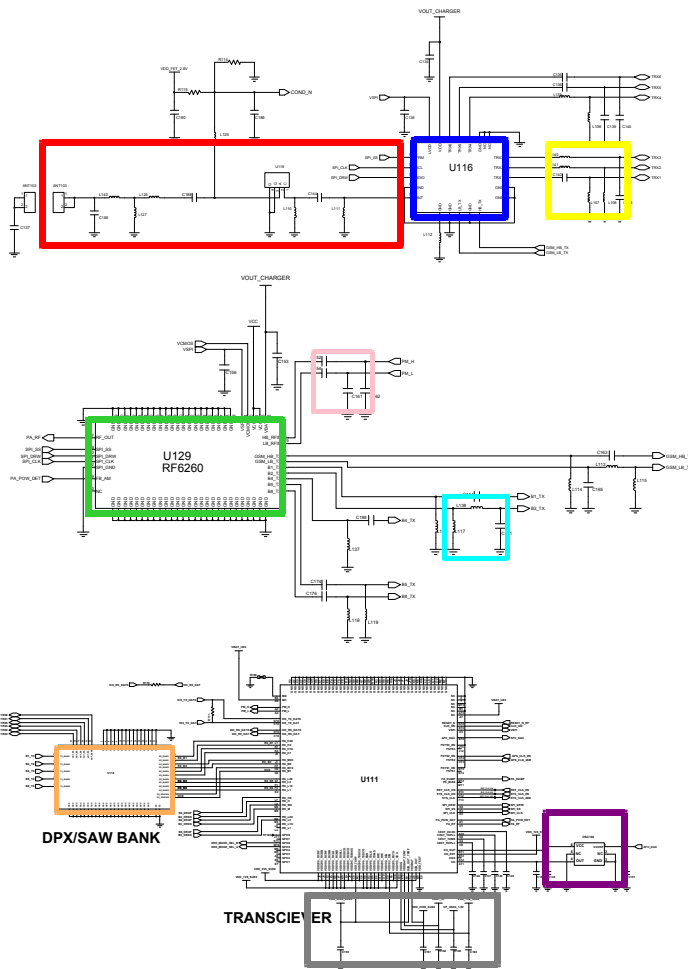
TRANSCIEVER DCDC



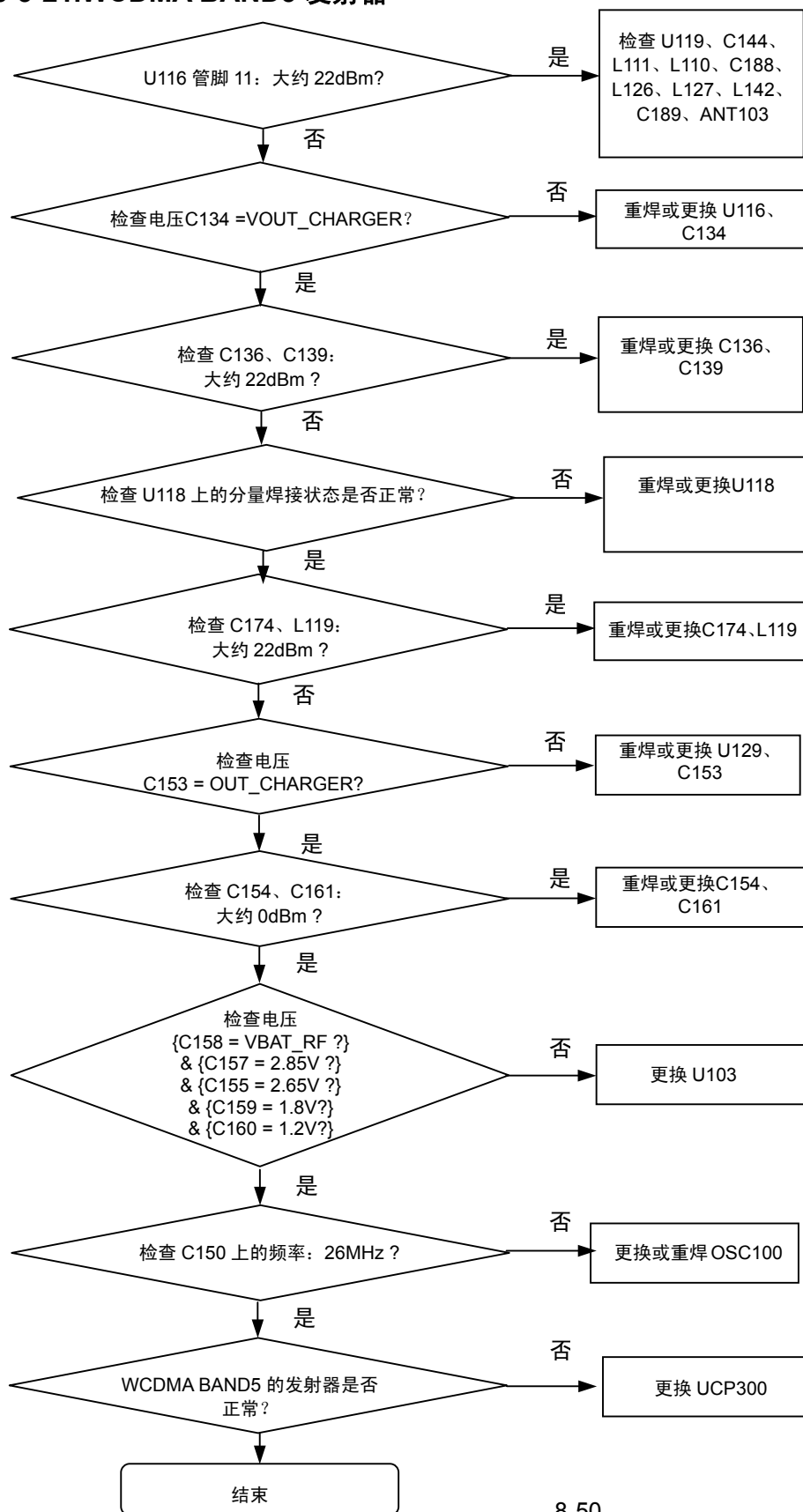
8-3-20.WCDMA BAND2 发射器



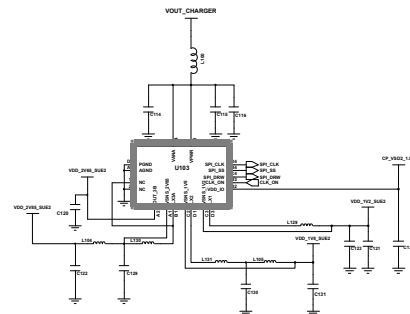
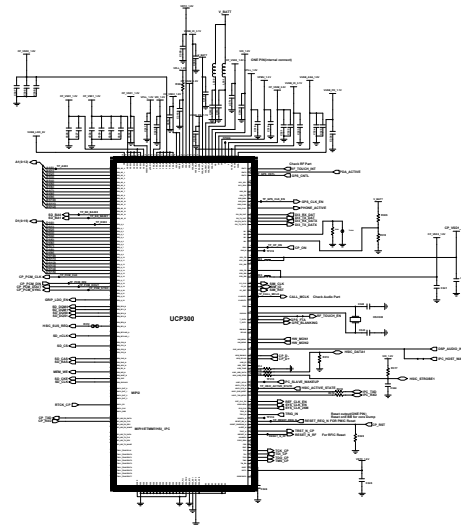
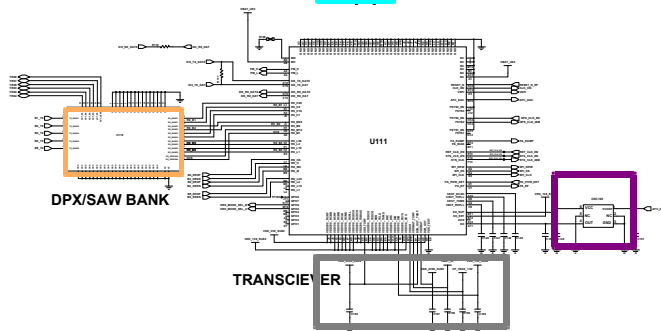
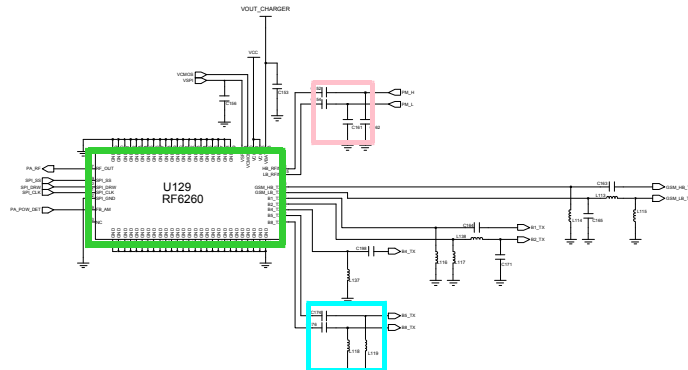
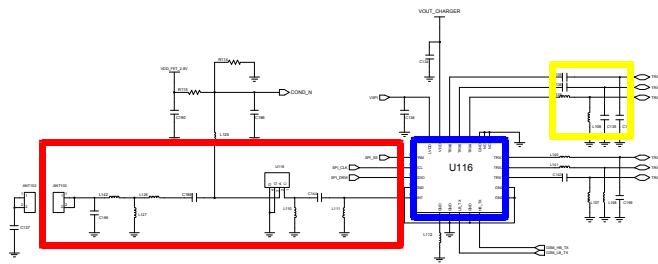
持续发射开通条件
 DAC 发射功率: 应用代码 14500
 WCDMA Band2 CH: 9880
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



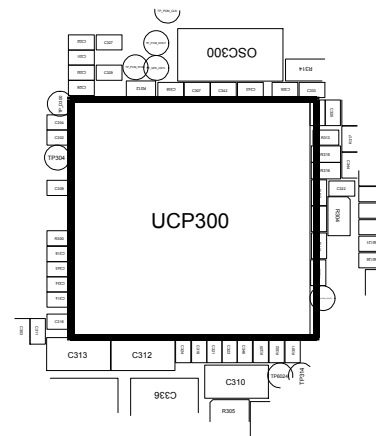
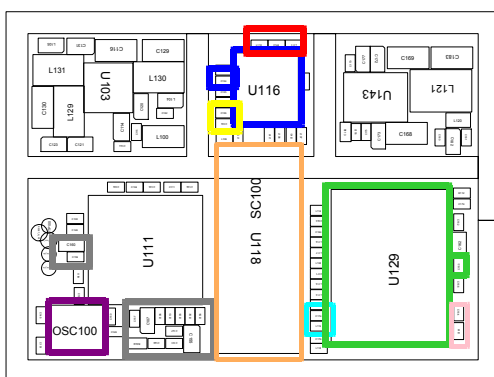
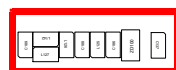
8-3-21.WCDMA BAND5 发射器



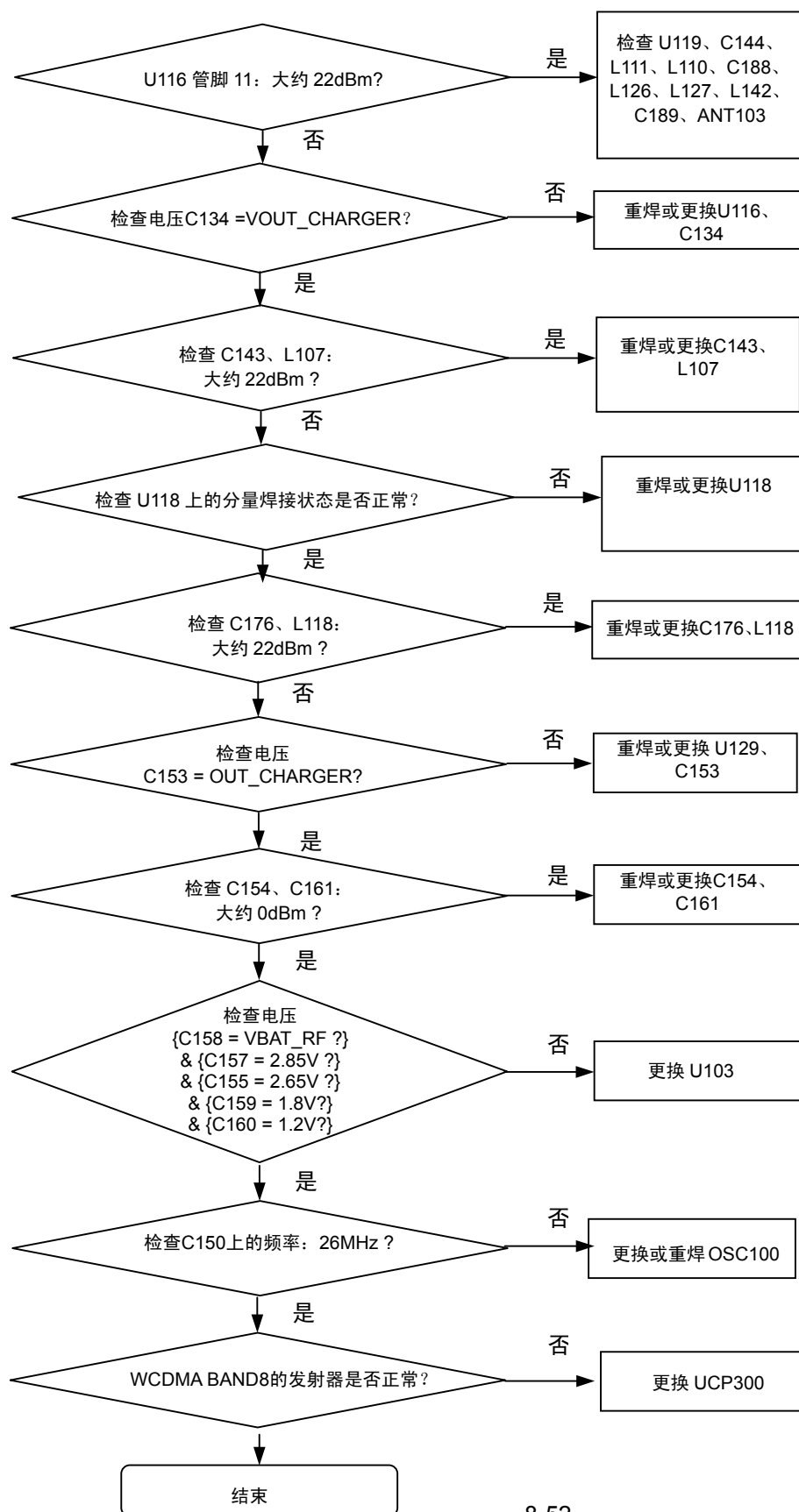
持续发射开通条件
 DAC 发射功率: 应用代码 14500
 WCDMA Band5 CH: 4408
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



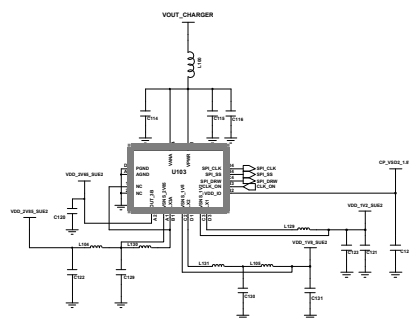
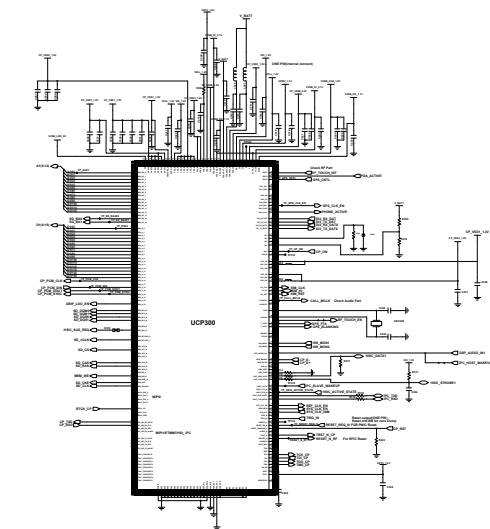
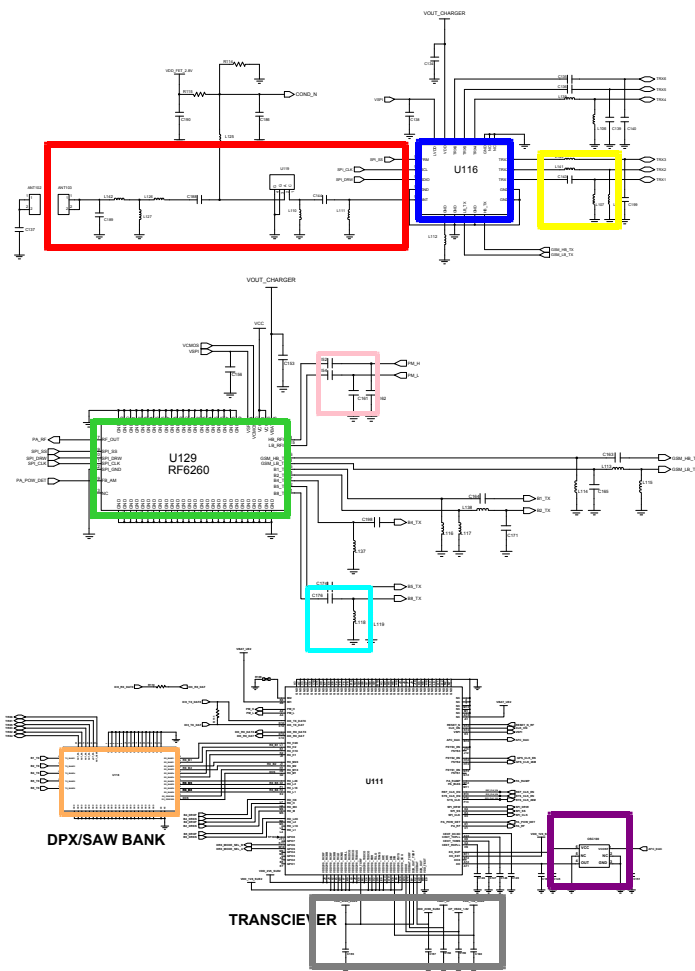
TRANSCIEVER DCDC



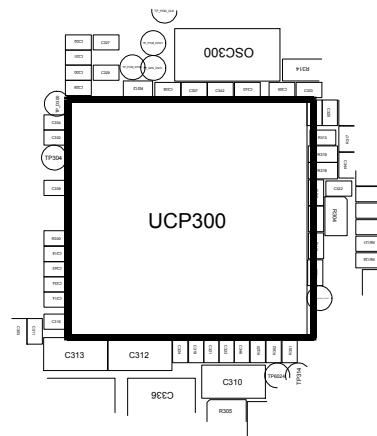
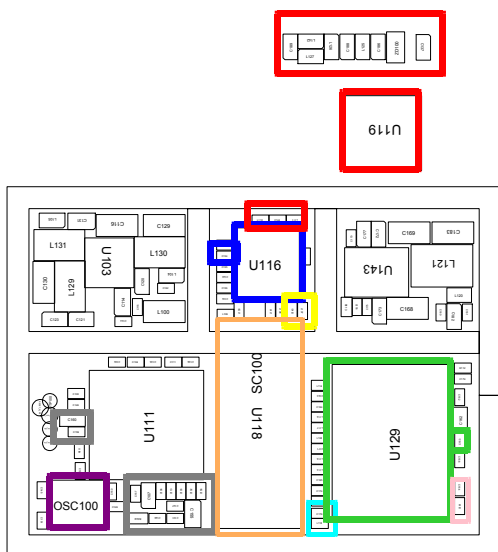
8-3-22.WCDMA BAND8 发射器



持续发射开通条件
 DAC 发射功率: 应用代码 14500
 WCDMA Band2 CH: 3013
 RBW: 100KHz
 VBW: 100KHz
 SPAN: 10MHz
 REF LEV.: 10dBm
 ATT.: 20dB



TRANSCEIVER DCDC



8-4.维修原理图

-NC 点（顶部视图）

: NC

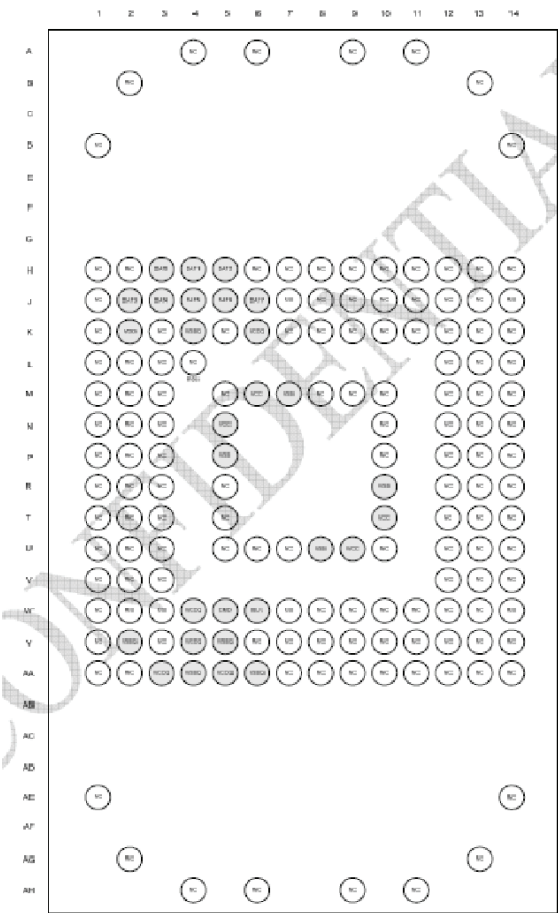
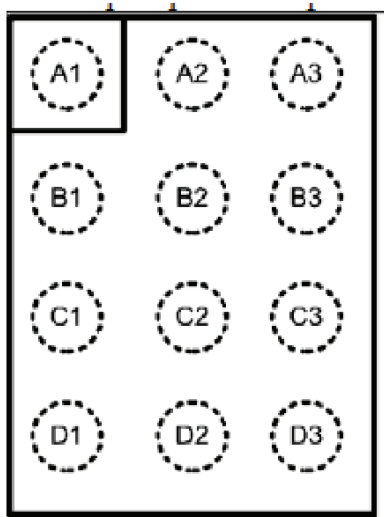


Figure 6- 169 balls - Ball Array (Top View)

U805

顶部视图

NC:D1



U1102

	1	2	3	4	5	6	7
A	RSVDL	VDDQ	MHL_D	MHL_DB	NC	NC	CBUS_ID
B	RSVDL	RSVDL	RSVDL	GND	AVCC33	USB_ID	RSVDL
C	CSCL	RSVDL	RSVDL	AVCC12	GND	V_SENSE	INT
D	CSDA	RSVDL	GND	IOVCC18	CVCC12	C2CA	RESET_N
E	DSDA	GND	IOVCC18	CVCC12	AVCC33	AVCC12	GND
F	RX2_P	GND	RPWR	DSCL	AVCC12	GND	HPD
G	RX2_N	RX1_P	RX1_N	RX0_P	RX0_N	RXC_P	RXC_N

Figure 3. Ball Diagram—HDMI Interface (Top View)

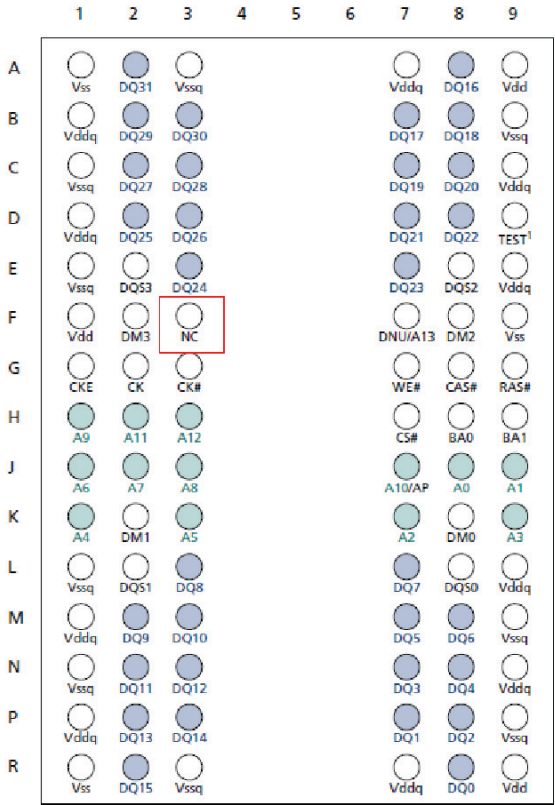
UME6000

134Ball FBGA										
	1	2	3	4	5	6	7	8	9	10
A	DNU	DNU	NB	NB	NB	NB	NB	NB	DNU	DNU
B	DNU	NC	NC	NB	VDD2	VDD1	DQ31	DQ29	DQ26	DNU
C	VDD1	VSS	ZQ1	NB	VSS	VSSQ	VDDQ	DQ25	VSSQ	VDDQ
D	VSS	VDD2	ZQ0	NB	VDDQ	DQ30	DQ27	DQ53	DQ53	VSSQ
E	VSSCA	CA9	CA8	NB	DQ28	DQ24	DM3	DQ15	VDDQ	VSSQ
F	VDDCA	CA6	CA7	NB	VSSQ	DQ11	DQ13	DQ14	DQ12	VDDQ
G	VDD2	CA5	VRef(CA)	NB	DQ51	DQ51	DQ10	DQ9	DQ8	VSSQ
H	VDDCA	VSS	CK	NB	DM1	VDDQ	NB	NB	NB	NB
J	VSSCA	NC	CK	NB	VSSQ	VDDQ	VDD2	VSS	VRef(DQ)	NB
K	CKE0	CKE1	NC	NB	DM0	VDDQ	NB	NB	NB	NB
L	CS0	CS1	NC	NB	DQ50	DQ50	DQ5	DQ6	DQ7	VSSQ
M	CA4	CA3	CA2	NB	VSSQ	DQ4	DQ2	DQ1	DQ3	VDDQ
N	VSSCA	VDDCA	CA1	NB	DQ19	DQ23	DM2	DQ0	VDDQ	VSSQ
P	VSS	VDD2	CA0	NB	VDDQ	DQ17	DQ20	DQ52	DQ52	VSSQ
R	VDD1	VSS	NC	NB	VSS	VSSQ	VDDQ	DQ22	VSSQ	VDDQ
T	DNU	NC	NC	NB	VDD2	VDD1	DQ16	DQ18	DQ21	DNU
U	DNU	DNU	NB	NB	NB	NB	NB	NB	DNU	DNU

[顶部视图]

UME300

顶部视图



UCP300

NC : 

(底部视图)

	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	
A	NC	MEM_A_9	MEM_BFC_LKO_1	MEM_SDC_LKO	MEM_A_8	MEM_BFC_0	MEM_A_1	MEM_A_2	MEM_A_3	MEM_A_4	MEM_AD_9	FCDP_RB_0	VPLL	VDD_VBAT_SD2	SD2_SW	SD2_SW	NC	A
B	MEM_A_9	MEM_A_2_3	MEM_BFC_LKO_0	MEM_CKE	MEM_RD_0	MEM_WR_0	VDD_IO1B	MEM_BC2_0	MEM_A_6	FWP	MEM_CAS_0	DSP_AUDI_O_IN1	x	VIO_12	x	x	VSS_SD2	B
C	MEM_A_1_0	MEM_A_2_2	x	x	x	MEM_A_5	x	x	MEM_A_7	x	x	x	x	VUSB_PD	VDD_VSD_2	SD2_FB	VDD_VBAT_SD1	C
D	MEM_A_1_1	MEM_A_2_1	x	VSS	MEM_AD_13	MEM_AD_12	MEM_AD_11	MEM_AD_7	VSS	MEM_AD_3	x	x	x	ANAMON	SD1_FB	x	SD1_SW	D
E	MEM_A_1_2	MEM_A_2_0	x	MEM_RAS_0	VDDCORE	VDDCORE_EBU	MEM_AD_8	VDDCORE	MEM_AD_6	MEM_AD_5	x	x	x	VSS	ISS	x	SD1_SW	E
F	MEM_A_1_3	MEM_BC1_0	VDD_IO1B	VSS	MEM_AD_15	x	x	x	MEM_AD_4	x	x	TRIG_B	x	VRTC	VDD_VBAT_PMU	VUSB_AN_A	VSS_SD1	F
G	MEM_A_1_4	MEM_BC1_0	MEM_A_1_9	MEM_BE0_0	MEM_BE1_0	x	MEM_AD_1	MEM_AD_V_0	MEM_AD_2	VSS	x	x	x	x	x	VPMU	VSIM	G
H	MEM_A_1_5	MEM_A_1_7	x	MEM_CS2_0	MEM_CS3_0	x	MEM_AD_0	MEM_AD_10	VDDCORE	RESERVE_D	x	M1	x	x	AGND	ON1	VUSB_IO	H
J	VDD_IO1B	MEM_A_1_6	MEM_A_1_8	MEM_CS0_0	MEM_CS1_0	MEM_AD_14	VSS	x	x	VSS	x	M2	M3	M4	RESET_P_WRDWN_N	ON2_N	VREF	J
K	MEM_WA1T_0	VDD_MIPI	x	VDDCORE	VSS	x	VDD_IO1B	EINT3	VDDCORE	VDD_IO1B	ETM11_T_RACECLK	x	ETM11_T_RACEPKT_4	TM_EN	REF_CLK_EN	RESET_B_N	ETM11_T_RACEPKT_7	K
L	F32K	VSS	x	VSS	MIPI_HSI_RX_RDY	x	MIPI_HSI_TX_FLG	VSS	ETM11_T_RACEPKT_5	VDD_MM_C	MMCH1_C_MD	x	ETM11_T_RACEPKT_6	ETM11_T_RACEPKT_2	x	ETM11_T_RACEPKT_3	ETM11_T_RACEPKT_1	L
M	OSC32K	VSS	MIPI_HSI_TX_RDY	MIPI_HSI_RX_FLG	MIPI_HSI_TX_WAKE	x	x	x	EINT1	EINT2	x	MMCH1_D_AT_3	MMCH1_D_AT_0	MMCH1_D_AT_2	MMCH1_C_1K	MMCH1_C_AT_1	ETM11_T_RACEPKT_0	M
N	VDD_RTC	USB_ID	x	VSS_USB	MIPI_HSI_RX_DATA	MIPI_HSI_TX_DATA	VDD_IO1B	I2C1_SCL	VPP	CLKOUT0	x	x	MMCH1_C_D	T_OUT0	x	I2S2_CLK_1	I2S2_CLK_0	N
P	USB_TSTUS	VBUS	x	VDD_USB_ANA	MIPI_HSI_RX_WAKE	SYS_CLK	VSS_PLL	CLKOUT2	I2C1_SDA	VSS	HW_MON_2	x	x	x	T_OUT1	I2S2_TX	I2S2_RX	P
R	USB_DM1NUS	VDD_USB1_O	x	x	x	VDD_PLL	x	x	VSS	VDDCORE	x	HW_MON_1	RESET2_N	NC	x	I2S2_WA1	I2S2_WA0	R
T	USB_DPLUS	USB_TUNE	VDD_USB_PD	CC_RST	CC_IO	VDD_SIM	DMINUS	VDD_DIG_RP	XRESET_N	D03_TX_D_ATX	D03_TX_D_AT	TCK	TRIG_IN	TRST_0	USIF1_TX_D_MSTR	USIF1_RX_D_MRST	USIF1_SCLK	T
U	NC	VSS	VDD_IO12	CC_CLK	HSIC_USB_DATA	HSIC_USB_STRB	DPLUS	ALERT_N	SYSCLKEN	D03_RX_D_ATX	D03_RX_D_AT	TMS	TDI	TDO	USIF1_RT_S_N	USIF1_CT_S_N	NC	U
	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	

U111

(底部视图)

	1	2	3	4	5	6	7	8	9	10	11	12	13	
A	NC	PM_H	GND	PM_L	GPO3	VDD1V8_TXDCO	GND	MI1	VDDBA_T	CEXT_D CXO	XO	XOX	NC	A
B	VDDBA_T_TXRF	VDD_TX RF	NB	NB	GPO2	NB	NB	MI2	D2B_OUT	AFC_DAT C	XO_SUP	XO_EXT	VDD1V8_SCU	B
C	PA_RF	D2B_OUT_TXRF	GND	CEXT_TXMS	VDD2V5_TXBIA S	VDD2V5_TXDCO	GND	GND	VDD1V8_DIG	GND	GND	DI3_TX_DAT	DI3_RX_DAT	C
D	PA_PO_W_DET	VDD1V8_FBR	SPI_CLK	SPI_DRW	VDD1V8_TXLO	NB	NB	CEXT_TXPLL	GND	VDD2V5_FSYS	NB	DI3_TX_DATX	DI3_RX_DATX	D
E	GND	RX_L1	V5PI	SPI_SS	GND	GND	GPO1	VDD1V8_TXMS	VDD_TEST	NB	NB	VDDIO	REF_CLK_EN	E
F	RX_L2	RX_L1X	VDD1V8_RCB	GND	GND	GND	VDD1V8_RCM3	GND	VDD1V2_DIG	GND	NB	SYS_CLK_EN	SYS_CLK	F
G	RX_L2X	RX_M1	GND	GND	VDD1V8_RCL	GND	GND	GND	GND	GND	NB	FSYS1	FSYS1_EN	G
H	RX_M2	RX_M1X	GND	GND	GND	GND	NB	NB	CEXT_RXPLL	GND	NB	FSYS2_EN	FSYS2	H
J	RX_M2X	RX_H1	GND	VDD2V5_RCB	VDD2V5_RCRF	GND	GND	VDD2V5_RXDCO	GND	GND	GPO8	FSYS3	FSYS3_EN	J
K	RX_H2	RX_H1X	NB	NB	GND	GND	NB	NB	VDD1V8_RXPLL	GND	GPO7	RESET_N	CLK_ON	K
L	RX_H2X	VDD1V8_RCRF	GND	RD_L1	RD_L1X	RD_H	RD_HX	GND	VDD1V8_RXDCO	VDD2V5_RFC	GND	GND	VDD1V8_RFC	L
M	NC	GND	RD_L2	RD_L2X	RD_M	RD_MX	GND	GPO4	GPO5	GPO6	PA_BIAS	PA_RAMP	NC	M
	1	2	3	4	5	6	7	8	9	10	11	12	13	

Figure 2 Ball Diagram PG-WFWLB-138-2 (Top View)

U143

顶部视图

